

Application of the Laplacian Salp Swarm Algorithm to Continuous Wind Farm Layout Optimization

Prince Solanki, Pulkit Dwivedi, Vanita Garg, and Vinay Shukla

Abstract—The Wind Farm Layout Optimization Problem (WFLOP) is a nonlinear constrained optimization problem in which turbine locations interact through wake-induced velocity deficits. This study employs the Laplacian Salp Swarm Algorithm (LX-SSA), previously introduced by Solanki and Deep in 2023 [30], for continuous WFLOP; LX-SSA is neither proposed nor redesigned in the present paper. The contribution is application-level: the published optimizer is coupled to a continuous turbine-coordinate representation, Jensen wake interactions, direction-dependent Weibull wind statistics, and explicit farm-boundary and minimum-spacing constraints. LX-SSA is compared with SSA, particle swarm optimization (PSO), differential evolution (DE), variable neighbourhood search (VNS) and a gradient-based multistart SLSQP solver on 68 benchmark cases (two wind data sets, three circular farms, 2–15 turbines). Each comparison uses 30 seed-paired runs at an equal budget of 6,030 objective evaluations and is analyzed with Friedman, Holm-corrected Wilcoxon signed-rank and effect-size statistics, together with feasibility rates and convergence curves. VNS performs best (average rank 1.60), followed by SSA, PSO and SLSQP; LX-SSA ranks fifth (3.97), ahead of DE. Against SSA, LX-SSA is never significantly better and is significantly worse in nine cases, so its Laplace follower step brings no equal-budget benefit. The salp-swarm algorithms reach feasible layouts in dense farms more reliably than PSO and DE, but less reliably than VNS and SLSQP. On the Horns Rev 1 offshore site, with its real turbine, wind climate and outline, the same ordering appears, no method improves on the installed layout within the budget, and only the constraint-aware SLSQP finds feasible layouts for the full 80-turbine farm. The algorithm ordering is insensitive to the power-curve model but sensitive to the wake model, and absolute energy values are strongly model-dependent. The conclusions are therefore limited to the stated benchmark, and no new algorithm, wake model or superiority over modern WFLO methods is claimed.

Index Terms—Laplacian Salp Swarm Algorithm (LX-SSA), Optimization, Renewable energy, Wake effect, Wind farm layout.

I. INTRODUCTION

Wind energy has emerged as a key component of sustainable energy systems. The efficient design of wind farms is governed

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by the Wind Farm Layout Optimization Problem (WFLOP), which aims to maximize energy production while minimizing wake-induced losses under spatial constraints. The nonlinear and highly coupled nature of wake interactions makes WFLOP a challenging optimization problem.

A wide range of optimization techniques have been applied to WFLOP, including Genetic Algorithms (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), Ant Colony Optimization (ACO), and hybrid approaches. While these methods have shown effectiveness, many suffer from issues such as premature convergence, sensitivity to parameter settings, and limited exploration capability in high-dimensional continuous search spaces.

Recently, swarm intelligence methods such as the Salp Swarm Algorithm (SSA) have gained attention due to their simplicity and low computational complexity. However, standard SSA often exhibits limited exploration capability and may become trapped in local optima, particularly in complex continuous optimization problems such as WFLOP.

Solanki and Deep [30] introduced the Laplacian Salp Swarm Algorithm (LX-SSA) in 2023 as an enhanced variant of SSA in which a Laplace-distribution-based position update is used to improve exploration and population diversity. The algorithmic design, motivation, benchmark-function validation, and original introduction of LX-SSA belong to that earlier publication. The present manuscript does *not* propose a new SSA variant. Instead, it investigates whether the already published LX-SSA can be effectively employed for continuous WFLOP, where each candidate solution represents turbine coordinates and must satisfy wake-dependent and geometric feasibility requirements.

Accordingly, the novelty claimed here is application-level rather than algorithm-design novelty. The standard WFLOP ingredients—Jensen’s wake model, a Weibull wind representation, and a benchmark turbine power model—are retained deliberately to permit comparison with the established WFLOP literature. The study focuses on how the previously developed LX-SSA behaves when coupled to this constrained continuous layout problem across multiple farm sizes and wind scenarios.

The contributions of the present application study are therefore summarized as follows:

- formulation of a continuous turbine-coordinate representation that permits the previously published LX-SSA [30] to be employed for WFLOP under explicit farm-boundary and minimum-spacing constraints;
- implementation of LX-SSA within the standard Jensen–Weibull WFLOP benchmark framework, without claim-

ing novelty in the wake, wind-resource, or turbine-power models themselves;

- a controlled comparison of LX-SSA with SSA, PSO, DE, VNS and a gradient-based multistart SLSQP solver on 68 benchmark cases at equal objective-evaluation budgets, with 30 seed-paired runs per method and case, full statistical analysis (Friedman and Holm-corrected Wilcoxon signed-rank tests, effect sizes), feasibility rates and convergence curves (Section VII);
- an application to the Horns Rev 1 offshore site with its real turbine, wind climate and outline, compared against the installed layout (Section VII-H); and
- a sensitivity analysis of the benchmark assumptions (power curve, wake model, minimum spacing and boundary handling; Section VIII).

This framing distinguishes the contribution of the present paper from the original LX-SSA development: the 2023 study introduced the optimizer, whereas the current study evaluates its applicability to WFLOP.

Renewable energy systems are increasingly being used as alternatives to conventional fossil fuel resources and climate change since they can decrease greenhouse gas (GHG) emissions, transmission losses, and transformation losses. Wind energy is a significant contributor to sustainable energy resources. Wind energy is typically harnessed through wind turbines, which convert the kinetic energy of moving air into mechanical energy and subsequently into electricity.

The efficient utilization of this resource leads to the Wind Farm Layout Optimization Problem (WFLOP), which involves determining the optimal spatial arrangement of wind turbines within a defined area to maximize energy production while considering constraints such as wake effect, turbine spacing, etc. The wake effect is a critical factor that must be carefully considered in the design of a wind farm layout. The wake produced by an upstream turbine leads to a reduction in wind speed, thus reducing the performance of downstream turbines. The wake effect can be modeled using generalized wake effect models that incorporate thrust and associated turbulence level. The wake spreads out downwind and its velocity deficit decays with distance. Packing more turbines into a fixed area raises installed capacity but also brings turbines closer together, which increases wake interaction and reduces the energy produced per turbine; minimum-spacing constraints are therefore imposed, and their value directly affects how many turbines can be placed (see Section VIII).

Nowadays, the production of wind energy is increasing rapidly, and it is likely to increase further in the coming years. The cost associated with generating wind energy poses a major challenge during the initial stage of wind farm development. One of the most effective strategies to mitigate these costs is through the optimization of the wind farm layout, which can significantly enhance energy capture while minimizing infrastructure and operational expenses. Over the years, researchers have used various optimization techniques and algorithms, including Genetic Algorithm (GA), Simulated Annealing (SA), Particle Swarm Optimization (PSO), Ant Colony Optimization (ACO), and Biogeography-Based Optimization (BBO) to address this problem.

Mosetti et al. [1] and Grady et al. [2] solved WFLOP by using a binary-coded GA to obtain the maximum energy output with the minimum installation costs. Ozturk and Norman [3] continued Mosetti's work on WFLOP with a different objective function and a heuristic approach that considered a continuous representation instead of a discrete one. The authors proposed a methodology for optimizing wind farm layouts with the objective of maximizing annual energy production (AEP), while considering key environmental and operational factors such as wind speed, wind direction, and wake interactions. Lackner and Elkinton [4] proposed an analytical model for optimizing offshore wind farms. The model's objective function was designed to maximize the annual power output of the wind farm, which is inherently dependent on turbine placement and wake effects. Castro et al. [5] used GA to design the optimal layout of the wind farms. The objective function aimed to maximize the farm's annual energy output, subject to constraints such as turbine spacing, wind speed, etc. Huang [6] used a distributed GA to find the optimal layout of wind turbines in large wind farms to obtain the maximum annual profit, which depends on factors such as energy output, capital cost, and operating cost. Elkinton et al. [7] employed a methodology for offshore wind farm layout optimization that used various optimization algorithms. To evaluate the effectiveness of their approach, the authors compared the performance of four well-known optimization algorithms: Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Simulated Annealing (SA). The objective of the optimization problem was to obtain the maximum energy output subject to some constraints. Bilbao and Alba [8] used SA to optimize the locations of wind turbines to maximize annual profit (as in Huang [6]). Rivas et al. [9] also used SA that performs three types of local search operations iteratively to find the optimal layout of the turbines in a large offshore wind farm. Emami and Nogreh [10] proposed a novel objective function that includes the construction cost of a wind farm to solve WFLOP using GA. The overarching goal was to maximize the annual energy yield while simultaneously minimizing the total construction cost. Sisbot et al. [11] used a multi-objective GA to optimize wind farm layout design. The algorithm maximized energy production while minimizing the installation cost of turbines. Kusiak and Song [12] developed a wind distribution-based turbine placement model that considered wake loss as a function of wind turbine locations and wind direction. This problem was reformulated into a bi-criteria optimization task aimed at simultaneously maximizing energy output and minimizing constraint violations. To address this, the authors employed a multi-objective Evolutionary Strategy (ES) to derive optimal solutions. Eroglu et al. [13], [14] proposed Ant Colony Optimization (ACO) and Particle Filtering (PF) approaches to find an optimal solution by iteratively improving the layout of the wind turbines to maximize the expected energy output while maintaining wake loss. Samorani [15] demonstrated the trade-off between maximizing power output and minimizing the wake effect within several turbines while solving WFLOP. Bansal and Farswan [16] employed Biogeography-based Optimization (BBO) to optimize the wind farm layout, proposing

a maximum count of turbines that should be placed in a wind farm to obtain maximum power with minimum wake loss. Ant Colony Optimization (ACO) has been employed to optimize the collector system topology of offshore wind farms by formulating the design problem as a multiple traveling salesman problem, resulting in reduced cable length and associated costs [27]. Ju and Liu [17] proposed two novel algorithms called Adaptive Genetic Algorithm (AGA) and Self-Informed Genetic Algorithm (SIGA) that incorporate the self-adaptive capability of individuals in a Conventional Genetic Algorithm (CGA) to solve WFLOP. Jin *et al.* [18] developed a power loss cost model for offshore wind farms that accounts for the wake effect on the wind turbine output and used an Adaptive Particle Swarm Optimization (APSO) algorithm to optimize the layout design. A comprehensive review of state of the art advances in WFLO and electrical system design, including cable connection scheme optimization, is provided in [28]. Dhiman *et al.* [19] proposed a wake management technique that uses lidar (light detection and ranging) simulations to detect the wake's location and intensity and then redirects the downstream turbines' yaw angle to reduce the wake effect. Nagpal *et al.* [20] proposed a local, continuous refinement approach to refine the layouts generated by a gradient-free heuristic algorithm called Distributive Genetic Algorithm (DGA). Particle Swarm Optimization (PSO) has been successfully employed for solving WFLO to reduce wake effects and improve overall power output [29]. Bai *et al.* [21] proposed an enhanced Adaptive Genetic Algorithm (AGA) for solving the optimal layout problem of wind turbines in a wind farm. The authors modeled the layout optimization task as a single-player reinforcement learning problem and integrated Monte-Carlo tree search into an evolutionary algorithm to enhance its exploitation capability. The proposed algorithm was then applied to optimize the layout of a recently approved wind farm project in New Jersey. Cazzaro and Pisinger [22] developed an advanced heuristic for solving WFLOP using the Variable Neighborhood Search (VNS) meta-heuristic. The heuristic considers the turbine spacing constraint and construction costs in the optimization process. The WFLO literature has also moved beyond classical population-based metaheuristics. Pseudo-gradient methods have been developed to obtain faster layout refinement for large design spaces [33]; three-dimensional Gaussian wake formulations have been coupled with layout optimization to improve wake representation [34]; machine-learning wake surrogates have been used for large-scale layout design [35]; CFD-based Kriging surrogates have been introduced to reduce the computational burden of higher-fidelity evaluations [36]; and recent work combines data-driven and physics-informed models for large-scale WFLO [37], [38]. These developments show that no single optimizer or wake model is universally dominant: current research balances wake-model fidelity, computational cost, scalability, and optimization robustness.

Against this background, the purpose of the present study is deliberately narrower than proposing a new WFLO paradigm. LX-SSA was already introduced and benchmarked as a continuous optimizer in 2023 [30]. Here it is *employed* in a standard continuous WFLO benchmark so that its behavior can

be examined under geometric constraints and wake-coupled objectives. Jensen's model and the benchmark Weibull/power formulation are retained primarily for comparability with the established literature; they are not presented as state-of-the-art wake physics or as novel modeling contributions.

II. SUITABILITY OF LX-SSA FOR WFLOP

This section motivates the use of the previously published LX-SSA [30]; it does not re-introduce LX-SSA as a new algorithm. Continuous WFLOP contains $2N$ spatial decision variables for N turbines, nonlinear wake coupling, and geometric feasibility constraints. Such features can produce many locally attractive layouts and make diversity preservation relevant to a population-based search.

In the 2023 LX-SSA, the standard SSA follower dynamics are supplemented by a Laplace-distribution-based candidate update and greedy retention [30], [31]. The heavy-tailed Laplace perturbation can generate both short and occasional longer moves, which provides a plausible mechanism for exploring spatially separated layout configurations. This is a *motivation for testing* LX-SSA on WFLOP rather than proof that it must outperform other algorithms.

The present experiments therefore ask an application question: when the already-developed LX-SSA is placed inside the same WFLOP model used for the SSA baseline, does the additional Laplace-based search step produce reproducible changes in the attained benchmark objective and wake loss? The comparison with SSA is treated as a mechanism-level baseline because the principal algorithmic distinction is the additional LX-SSA update. PSO, DE, VNS and a gradient-based solver are added as further baselines, all compared at equal objective-evaluation budgets (Section VII); pseudo-gradient, surrogate and data-driven WFLO methods remain literature context.

The remainder of the paper presents the benchmark model, summarizes the previously published optimizer only to the extent needed for reproducibility, defines its WFLOP mapping and constraint handling, reports the experimental results, and finally states the limitations of the evidence.

III. PROBLEM MODELING

Certain definitions and presumptions are necessary to design a wind farm and determine the optimal placements of turbines, which are as follows:

- 1) **Installed Capacity and Prescribed Turbine Count:** For identical turbines, nameplate capacity scales linearly with the number of installed units, but the *expected farm power does not* because wake interactions depend on the layout. In each optimization run considered here, the turbine count N is prescribed in advance and LX-SSA optimizes only the $2N$ planar coordinates. Results across several values of N therefore constitute a turbine-count sweep, not simultaneous optimization of N and turbine locations.
- 2) **Turbine Location Representation:** The location of each turbine is represented in 2-dimensional space

(ρ, σ) , and the magnitude of its location vector is computed as $\sqrt{\rho^2 + \sigma^2}$.

- 3) **Uniformity of Turbines:** All turbines should be uniform in terms of internal quality (power curve, capacity) and external quality (design, model, hub height).
- 4) **Wind Speed Modeling:** To characterize wind speed variability, the Weibull distribution is commonly used to model its probability density function (PDF). For a specific location, height, and wind direction, the wind speed s is assumed to follow a Weibull distribution:

$$p_s(s; \zeta, \psi) = \frac{\zeta}{\psi} \left(\frac{s}{\psi} \right)^{\zeta-1} \exp \left(- \left(\frac{s}{\psi} \right)^{\zeta} \right),$$

where ζ corresponds to the shape parameter, ψ is the scale parameter, and $p_s(\cdot)$ represents the PDF.

- 5) **Directional Functions:** Wind speed (s) is dependent on the wind direction θ , expressed as $s = s(\theta)$. Similarly, both the shape parameter (ζ) and the scale parameter (ψ) vary as functions of wind direction, i.e., $\zeta = \zeta(\theta)$, $\psi = \psi(\theta)$, $0^\circ \leq \theta \leq 360^\circ$. Therefore, θ is an important parameter in WFLOP. Figure 1 illustrates a wind farm with four representative wind directions, with $\theta = 0^\circ$ representing the east, $\theta = 90^\circ$ representing the north, $\theta = 180^\circ$ representing the west, and $\theta = 270^\circ$ representing the south.
- 6) **Minimum Turbine Spacing:** To preserve comparability with the legacy benchmark studies used in the numerical section, the minimum center-to-center spacing is set to $4D = 8R$, where $D = 2R$ is the rotor diameter:

$$(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2.$$

This value is a *benchmark lower bound* retained for comparison with the legacy WFLOP literature [12], [16], not a universal engineering recommendation. Project-specific spacing requirements can be larger and direction-dependent; therefore conclusions about feasible turbine density are conditional on the present $4D$ assumption. Section VIII-B quantifies this dependence with a $4D/5D/6D$ sensitivity study of geometric capacity, attained objective and wake loss.

- 7) **Wind Farm Shape:** WFLOP focuses on determining the optimal spatial placement of wind turbines within a specified area. The shape of the wind farm can have an impact on the optimal layout. In WFLOP, the shape can be circular, rectangular, or any other feasible shape. For this study, a circular wind farm is chosen for consistency with the legacy benchmark studies [12], [16]; a circular domain also has no preferred orientation, so the boundary does not favor any wind direction. No claim is made that a circular site is superior in land use to other shapes.
- 8) **Farm Boundary Constraint:** To ensure that all the wind turbines are placed within the designated wind farm area, the location of each turbine $T_i(\rho_i, \sigma_i)$ must adhere to the following spatial constraint:

$$\rho_i^2 + \sigma_i^2 \leq r^2,$$

where r is the wind farm radius. This study examines three cases with radii of 500 m, 750 m, and 1000 m.

- 9) **Search Space:** In WFLOP, the search space is defined by the boundary of the wind farm, which is circular in shape as shown in Figure 1. Since the search space is continuous, there is no need to use a grid system to locate a wind turbine.

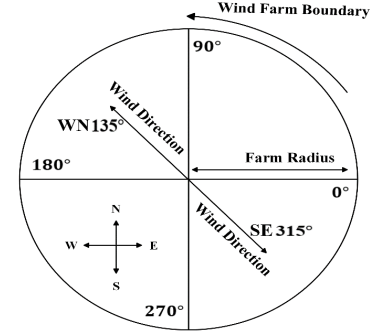


Fig. 1: A circular wind farm

- 10) **WFLOP Mathematical Model:** The mathematical model of WFLOP contains two parts: the wake effect model and the power output model. The wake effect model captures the reduction in power generation experienced by downstream turbines due to the wake induced by upstream turbines. Jensen's wake model [23], [24] is a widely used model for solving WFLOP. The power output model used in WFLOP is considered from Kusiak and Song [12].
- 11) **Objective:** This study aims to maximize the power generation by minimizing wake-induced losses, while satisfying two constraints obtained from assumptions (6) and (8).

A. The Wake Effect Model

The wake effect can lead to a substantial reduction in the wind speed experienced by downstream turbines as a result of momentum extraction and wake expansion. Therefore, accounting for wake losses is a critical aspect of wind farm layout design. Jensen's wake model [23], [24] is retained here because it is computationally inexpensive and has been used extensively in the legacy WFLOP benchmark literature against which the present results are compared. This choice is made for *benchmark comparability*, not because Jensen is the most accurate contemporary wake model. Gaussian and three-dimensional wake formulations can represent the velocity field more realistically and have been used successfully in WFLO studies [34], [39]; consequently, the numerical conclusions below are model-dependent.

When a uniform wind comes into contact with a turbine, it creates turbulence, and a wake expands linearly downstream, causing a reduction in wind speed, i.e., from upstream wind speed s_{up} to downstream wind speed s_{down} . Figure 2 illustrates the wake model of WFLOP.

Let s_{up} represent the wind speed before encountering the turbine, s_{down} represent the downstream wind speed, K be the

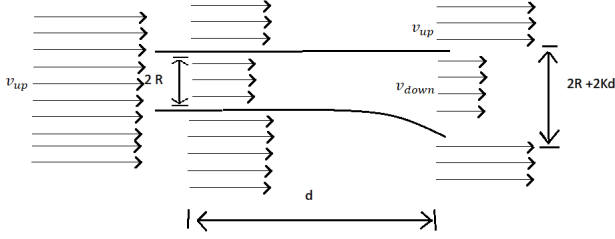


Fig. 2: Wake model representation for wind turbine layout

wake spreading constant, d be the inter-turbine distance, and R be the rotor radius of the turbine.

The velocity deficit induced at turbine j resulting from the wake generated by turbine i is given by:

$$\delta s_{ij} = 1 - (s_{down}/s_{up}) = \frac{1 - \sqrt{1 - C_T}}{\left(1 + \frac{K d_{ij}}{R}\right)^2}$$

where C_T denotes the turbine thrust coefficient, and d_{ij} represents the inter-turbine distance between turbines i and j .

If a turbine is influenced by the wakes of multiple upstream turbines, the total velocity deficit for that turbine is the superposition of the velocity deficits due to each wake. Assuming turbine i is influenced by the wakes of N turbines, then the overall deficit can be computed as follows:

$$\delta s_i = \sqrt{\sum_{\substack{j=1 \\ j \neq i}}^N (\delta s_{ij})^2}. \quad (1)$$

Each turbine rotor is positioned perpendicular to the prevailing wind direction θ to ensure maximum energy capture. When wind passes downstream of a turbine located at (ρ, σ) , the resulting wake can be idealized as an imaginary half cone. Figure 3 indicates this half cone generated by a turbine positioned at (ρ, σ) .

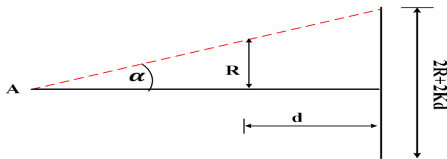


Fig. 3: Geometric representation of the wake expansion angle α

where A is the imaginary vertex of the half cone representing the wake region. The geometry of the cone is governed by the wake spreading constant K , α ($0 \leq \alpha \leq \pi/2$) is the half-cone angle, i.e., the angle between the central axis and the side of the cone. The value of α is computed as $\alpha = \arctan(K)$. This imaginary half cone is used to model the influence of wake effects from one turbine on others within the wind farm.

Lemma 1. For a specified wind direction θ , the angle β_{ij} , where $0 \leq \beta_{ij} \leq \pi$, represents the angle between the two vectors extending from point A to turbines i and j and it can be computed using the following expression:

$$\beta_{ij} = \cos^{-1} \left(\frac{u_{ij}}{\sqrt{u_{ij}^2 + v_{ij}^2}} \right),$$

with

$$\begin{aligned} u_{ij} &= (\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta + \frac{R}{K}, \\ v_{ij} &= -(\rho_i - \rho_j) \sin \theta + (\sigma_i - \sigma_j) \cos \theta, \end{aligned}$$

where $\frac{R}{K}$ denotes the distance from the rotor center to the virtual vertex A of the wake cone.

Lemma 2. If turbine i is positioned within the wake generated by turbine j , the distance between these, along the wind direction θ is given by:

$$d_{ij} = |(\rho_i - \rho_j) \cos \theta + (\sigma_i - \sigma_j) \sin \theta|.$$

The equation (1) is then expressed by restricting the summation to those turbines j where $j \neq i$ and $\beta_{ij} < \alpha$:

$$\delta s_i = \sqrt{\sum_{\substack{j=1 \\ j \neq i, \beta_{ij} < \alpha}}^N (\delta s_{ij})^2}. \quad (2)$$

Equation (2) indicates that δs_i is a function of both the wind direction θ and the turbine i 's location (i.e., (ρ_i, σ_i)).

The transformation of the Weibull parameter is made explicit as follows. For a fixed directional bin θ , let the free-stream speed be

$$S_\theta \sim \text{Weibull}(\zeta(\theta), \psi(\theta)).$$

Under the present Jensen benchmark, the combined deficit $\delta s_i(\theta)$ is treated as a deterministic multiplicative reduction within that directional bin, so the local inflow at turbine i is

$$S_{i,\theta} = [1 - \delta s_i(\theta)] S_\theta.$$

For a positive constant a , if $S \sim \text{Weibull}(\zeta, \psi)$, then $aS \sim \text{Weibull}(\zeta, a\psi)$. Hence the shape parameter is unchanged while the scale parameter becomes

$$\psi_i(\theta) = \psi(\theta) [1 - \delta s_i(\theta)], \quad i = 1, 2, \dots, N. \quad (3)$$

This scaling is physically consistent with the *specific simplifying assumption* that the wake model acts as a deterministic speed multiplier for a given direction bin. It should not be interpreted as a general turbulence model; more detailed wake/turbulence models may alter the full wind-speed distribution rather than only its scale.

B. The Power Model

The power model follows the simplified Kusiak–Song benchmark used by the comparison studies [12]. It is a computational surrogate for a manufacturer power curve rather than a claim of high-fidelity turbine aerodynamics:

$$f(s) = \begin{cases} 0, & s < s_{\text{cut-in}}, \\ \lambda' s + \eta, & s_{\text{cut-in}} \leq s \leq s_{\text{rated}}, \\ P_{\text{rated}}, & s > s_{\text{rated}}. \end{cases} \quad (4)$$

Here $s_{\text{cut-in}}$ and s_{rated} are the benchmark cut-in and rated speeds, while λ' and η define the linear segment. The legacy benchmark does not explicitly model a cut-out shutdown in the subsequent analytical integral; the present manuscript therefore uses the model consistently as written above. This simplification improves comparability with prior WFLOP results but can introduce error relative to a turbine-specific nonlinear power curve. Rather than assuming this error is negligible, Section VIII-A re-evaluates the stored layouts with a nonlinear (cubic) power curve, with and without a 25 m/s cut-out, and reports the resulting change in expected power and in the algorithm ordering.

For a turbine positioned at (ρ, σ) and wind direction θ , the expected power output is given by:

$$\begin{aligned} E(P_i) &= \int_0^{360} P_\theta(\theta) \cdot E(P_i, \theta) d\theta \\ &= \int_0^{360} P_\theta(\theta) \cdot \int_0^\infty f(s) \cdot p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta. \end{aligned} \quad (5)$$

Thus, the optimization problem is formulated with the objective of maximizing the total power output of the wind farm, subject to the spatial and operational constraints defined by assumptions (6) and (8). The mathematical formulation is:

$$\begin{aligned} &\text{Maximize} \quad \sum_{i=1}^N E(P_i) \\ &\text{subject to} \quad (\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2 \geq 64R^2, \\ &\quad \rho_i^2 + \sigma_i^2 \leq r^2, \quad i, j = 1, 2, \dots, N, \quad i \neq j \end{aligned}$$

where $E(P_i)$ corresponds to the expected power output of the i^{th} turbine under the adopted benchmark model. The piecewise-linear representation in (4) is retained for consistency with the benchmark literature, not because its approximation error is assumed negligible. Incorporating Equation (4) and the Weibull model, Equation (5) becomes:

$$\begin{aligned} E(P_i) &= \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s, \zeta(\theta), \psi_i(\theta)) ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \int_0^\infty \frac{f(s) \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds d\theta \\ &= \int_0^{360} p_\theta(\theta) \left[\lambda' \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{s \zeta(\theta)}{\psi_i(\theta)} \left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)-1} e^{-\left[\frac{s}{\psi_i(\theta)} \right]^{\zeta(\theta)}} ds \right. \\ &\quad \left. + \eta \int_{s_{\text{cut-in}}}^{s_{\text{rated}}} \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right. \\ &\quad \left. + P_{\text{rated}} \int_{s_{\text{rated}}}^\infty \frac{\zeta(\theta)}{\psi_i(\theta)} \left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)-1} e^{-\left(\frac{s}{\psi_i(\theta)} \right)^{\zeta(\theta)}} ds \right] d\theta. \end{aligned}$$

Wind direction and speed are discretized into $N_\theta + 1$ and $N_s + 1$ bins, respectively, using equal bin widths to approximate the integration part using the Riemann sum [25]. The discretization approach for estimating expected power output was described by Kusiak and Song [12].

Let wind direction bins be defined by $\theta_0 = 0^\circ, \theta_1, \dots, \theta_{N_\theta+1} = 360^\circ$ and wind speed bins between $s_{\text{cut-in}}$ and s_{rated} be $s_0 = s_{\text{cut-in}}, s_1, \dots, s_{N_s+1} = s_{\text{rated}}$.

Using the Riemann sum approximation [25], the expected power output of the i^{th} turbine is estimated as:

$$\begin{aligned} E(P_i) &= \lambda' \sum_{j=1}^{N_s+1} \left(\frac{s_{j-1} + s_j}{2} \right) \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{j-1}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} - e^{-\left(\frac{s_j}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \right] \\ &\quad + P_{\text{rated}} \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} e^{-\left(\frac{s_{\text{rated}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \\ &\quad + \eta \sum_{l=1}^{N_\theta+1} (\theta_l - \theta_{l-1}) \omega_{l-1} \\ &\quad \left[e^{-\left(\frac{s_{\text{cut-in}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} - e^{-\left(\frac{s_{\text{rated}}}{\psi_i(\frac{\theta_l + \theta_{l-1}}{2})} \right)^{\zeta(\frac{\theta_l + \theta_{l-1}}{2})}} \right]. \end{aligned}$$

Here, N_s and N_θ denote the count of intervals for wind speed and wind direction, respectively, and ω_{l-1} is the wind blowing probability for the $(l-1)$ -th wind direction bin.

a) *Interpretation of the reported benchmark objective.:*

The numerical tables retain the historical discretization convention of the adopted WFLOP benchmark, including the directional-bin-width factor $(\theta_l - \theta_{l-1})$. The resulting tabulated “power” values are therefore best interpreted as *benchmark objective values under that legacy convention* for comparison with prior studies, rather than as instantaneous electrical power that must be bounded by $N P_{\text{rated}}$. Because all 24 bins have the same width of 15° and $\sum_l \omega_{l-1} = 1$, the reported objective equals 15 times the expected farm power in kW. For example, the free-stream (ideal) objective of a single turbine is 14045.74 for Wind Data Set I, which corresponds to an expected power of 936.38 kW, i.e., a capacity factor of 0.624 and an annual energy production (AEP) of $936.38 \times 8.76 = 8202.7$ MWh. In general, $\text{AEP [MWh]} = (\text{objective}/15) \times 8.76$. This conversion was confirmed with the independent evaluator described in Section VI-E. This distinction is important when comparing the benchmark numbers with physical nameplate capacity.

C. Constraint Handling Mechanism

The WFLOP search must satisfy the circular farm boundary and the minimum inter-turbine spacing. The implementation used for all reported runs handles both constraints with two explicit operations: a box repair of the decision variables and a large additive penalty in the minimized objective. No other repair or displacement operator is applied.

1) *Box Repair:* Every coordinate of every candidate is clipped to the bounding square of the farm after each position update,

$$\begin{aligned} \rho_i &\leftarrow \min\{\max\{\rho_i, -r\}, r\}, \\ \sigma_i &\leftarrow \min\{\max\{\sigma_i, -r\}, r\}. \end{aligned}$$

This guarantees bounded decision variables but not the circular constraint $\rho_i^2 + \sigma_i^2 \leq r^2$, which is enforced through the penalty below.

2) *Violation Measures*: The benchmark constraints are $d_{ij} = \sqrt{(\rho_i - \rho_j)^2 + (\sigma_i - \sigma_j)^2} \geq d_{\min} = 8R$ for all $i \neq j$ and $\rho_i^2 + \sigma_i^2 \leq r^2$ for all i . Their violations are measured by

$$g_{ij}^s = d_{\min} - d_{ij}, \quad g_i^b = \rho_i^2 + \sigma_i^2 - r^2, \quad (6)$$

and a layout is feasible if and only if $g_{ij}^s \leq 0$ and $g_i^b \leq 0$ for all i and j .

3) *Penalized Objective*: All optimizers minimize the penalized wake loss

$$F_p = \left(NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \lambda g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \lambda g_{ij}^s)^2, \quad \lambda = 10^{10}, \quad (7)$$

where NE_{ideal} is the wake-free value of the objective, so the first term is the wake loss reported in the tables. Any violation larger than about 10^{-7} (m or m^2) makes the penalty exceed the largest attainable wake loss, so comparing F_p values implements feasibility-first selection:

- any feasible candidate is preferred to a (non-negligibly) infeasible one;
- between two feasible candidates, the one with the smaller wake loss (larger objective) is preferred;
- between two infeasible candidates, the one with the smaller sum of squared penalty terms is preferred.

Box repair is applied after every position update and F_p is used in every comparison, sort and greedy selection of Algorithms 1 and 2. The same objective, including the penalty, is used for SSA, LX-SSA, PSO and DE. Section VIII-C shows that a radial projection onto the circle, used instead of box clipping, changes the attained objective values, so the boundary-handling rule is part of the experimental protocol.

IV. PREVIOUSLY DEVELOPED LAPLACIAN SALP SWARM ALGORITHM (LX-SSA)

LX-SSA was introduced by Solanki and Deep in 2023 [30]; the algorithmic novelty belongs to that publication. The present paper only summarizes the operators needed to understand and reproduce the WFLOP application. Consistent with the published SSA/LX-SSA implementation, the sorted population is divided into a leader subgroup (first half) and a follower subgroup (second half), while the best solution found so far is retained as the food source H [26], [30]. LX-SSA keeps these SSA dynamics and supplements the follower update with a Laplace-based candidate and greedy retention [30], [31].

For a leader salp $i \leq N_p/2$, the j th coordinate is updated around the food source as

$$\rho_j^i = \begin{cases} H_j + r_1((ub_j - lb_j)r_2 + lb_j), & r_3 \geq 0.5, \\ H_j - r_1((ub_j - lb_j)r_2 + lb_j), & r_3 < 0.5, \end{cases} \quad (8)$$

where $r_2, r_3 \sim U(0, 1)$ and

$$r_1 = 2 \exp \left[- \left(\frac{4l}{L} \right)^2 \right],$$

with iteration l and maximum iteration L . For follower salps $i > N_p/2$, the standard SSA position is formed from the current and preceding salps:

$$\rho_j^i = \frac{1}{2} (\rho_j^i + \rho_j^{i-1}). \quad (9)$$

LX-SSA then forms an additional follower candidate

$$y_j^i = x_j^i + \gamma(z, \phi, \chi) (H_j - x_j^i), \quad (10)$$

where x_j^i is the current follower coordinate and

$$\gamma(z, \phi, \chi) = \begin{cases} \phi + \chi \ln(2z), & z \leq 0.5, \\ \phi - \chi \ln(2(1-z)), & z > 0.5, \end{cases} \quad (11)$$

with $z \sim U(0, 1)$, which is the inverse-distribution-function sample of a Laplace variable with location ϕ and scale χ (mean ϕ , variance $2\chi^2$). The location parameter ϕ centers the Laplace variate, whereas the scale parameter $\chi > 0$ controls its dispersion and therefore the perturbation magnitude. Smaller χ concentrates moves near the center; larger χ increases the probability of longer exploratory moves. All experiments use $\phi = 0$ and $\chi = 1$, so that γ is symmetric about zero with $P(|\gamma| > 1) = e^{-1} \approx 0.37$: most follower candidates move towards H ($\gamma > 0$) or away from it ($\gamma < 0$) by a fraction of the current distance, while a heavy tail occasionally produces moves beyond H or well away from it. These values are fixed for every WFLOP case and are not tuned to WFLOP; no WFLOP-specific sensitivity analysis of ϕ and χ was carried out. Because each follower evaluates both candidates before the population is re-evaluated, one LX-SSA iteration costs $N_p/2 \times 2 + N_p = 2N_p$ objective calls, twice the N_p calls of an SSA iteration; comparisons at equal numbers of calls are therefore reported in Section VII.

Algorithm 1 follows the published leader/follower logic and avoids the earlier ambiguity in which a follower update appeared inside the leader branch. It also avoids hard-coding a minimization inequality; “better” is determined by the objective sense and, for WFLOP, by the feasibility rules in Section III-C.

The comparison between SSA and LX-SSA in the WFLOP experiments can therefore be interpreted as a mechanism-level comparison (an ablation) of the baseline SSA dynamics against the additional Laplace-based follower candidate.

V. APPLICATION OF LX-SSA TO THE WIND FARM LAYOUT OPTIMIZATION PROBLEM (WFLOP)

The published LX-SSA [30] is employed without algorithmic redesign. For a prescribed number N of turbines, candidate j is represented by

$$\mathbf{x}_j = (\rho_j^1, \sigma_j^1, \rho_j^2, \sigma_j^2, \dots, \rho_j^N, \sigma_j^N) \in \mathbb{R}^{2N}.$$

The optimizer therefore searches only the turbine coordinates; N is fixed during a run. The objective and constraint calculations are the WFLOP model described above.

Algorithm 1 Previously developed LX-SSA search logic [30]

```

1: Initialize and evaluate the salp population; sort it and set
   the best solution as food source  $H$ .
2: while the termination criterion is not met do
3:   Update  $r_1$ .
4:   for salp  $i = 1, \dots, N_p$  do
5:     if  $i \leq N_p/2$  then
6:       Update leader  $i$  using (8).
7:     else
8:       Form the standard follower candidate using (9).
9:       Draw  $\gamma(z, \phi, \chi)$  using (11).
10:      Form the Laplace-based follower candidate using
        (10).
11:      Evaluate both follower candidates and retain
        the better one.
12:    end if
13:  end for
14:  Enforce variable bounds, re-evaluate the population,
   sort, and update  $H$ .
15: end while
16: return the best solution  $H$ .

```

Algorithm 2 makes the implementation sequence explicit. In particular, the candidate coordinates are updated first, box repair is applied, wake deficits and Weibull-scale adjustments are recomputed, the constraint violation is measured, and all comparisons use the penalized objective F_p , which implements feasibility-first selection (Section III-C). This ordering is essential because the objective depends on the layout-dependent wake field.

No claim is made that this mapping changes the mathematical definition of LX-SSA. Its contribution is the explicit integration of the existing optimizer with the continuous WFLOP representation and constraints.

VI. EXPERIMENTAL SETUP

A. Test Problems

Two wind data sets are used (Table I). *Wind Data Set I* has fixed Weibull parameters ($\zeta = 2$, $\psi = 13$) in all directions and a strongly directional blowing probability: $\omega_0 = \omega_{23} = 0$, $\omega_5 = 0.2$, $\omega_6 = 0.6$, and $\omega_l = 0.01$ for all other intervals. *Wind Data Set II* keeps $\zeta = 2$ but lets the scale parameter vary with direction, from $\psi = 2.6$ (255–270°) to $\psi = 10$ (180–195°), with a broader directional distribution. Wind direction is discretized into $N_\theta = 24$ bins of 15°, and the speed range from $s_{\text{cut-in}}$ to s_{rated} into $N_s = 21$ intervals of 0.5 m/s.

Each data set is combined with three circular farms of radius $r = 500, 750$ and 1000 m. The turbine count is prescribed for each run: $N = 2, \dots, 10$ for 500 m, $N = 2, \dots, 12$ for 750 m and $N = 2, \dots, 15$ for 1000 m. This gives $2 \times (9 + 11 + 14) = 68$ test cases. The turbine and wake parameters are

- rotor radius $R = 38.5$ m and minimum spacing $4D = 8R = 308$ m;
- cut-in speed $s_{\text{cut-in}} = 3.5$ m/s, rated speed $s_{\text{rated}} = 14$ m/s and rated power $P_{\text{rated}} = 1500$ kW;

Algorithm 2 Application of LX-SSA to WFLOP for a prescribed turbine count N

Require: Farm radius r , prescribed N , wind data, turbine/wake parameters, population size N_p , maximum iterations T , ϕ , χ

```

1: Set dimension  $d = 2N$ , bounds  $[-r, r]^d$ , and draw  $N_p$ 
   coordinate vectors uniformly in the bounds.
2: for each candidate do
3:   For each wind-direction bin, compute wake deficits
      $\delta s_i(\theta)$ .
4:   Set  $\psi_i(\theta)$  using (3) and compute  $E(P_i)$ .
5:   Compute the violations (6) and the penalized objective
      $F_p$  from (7).
6: end for
7: Sort the population by  $F_p$  and set the best candidate as
   food source  $H$ .
8: for  $t = 1$  to  $T$  do
9:   Generate the leader and follower candidates of Algo-
     rithm 1, applying box repair to each; followers keep the
     better of their two candidates by  $F_p$ .
10:  Apply box repair, evaluate  $F_p$  for the new population
    and sort it.
11:  If the best new candidate has a smaller  $F_p$  than  $H$ ,
    replace  $H$ .
12: end for
13: return  $H$  and its turbine coordinates, objective value,
    wake loss and feasibility status.

```

- linear power-curve parameters $\lambda' = 140.86$ and $\eta = -500$;
- thrust coefficient $C_T = 0.8$ and wake spreading constant $K = 0.075$;
- constraint penalty $\lambda = 10^{10}$ in (7).

Every run optimizes the $2N$ turbine coordinates for its prescribed N ; varying N is a parameter sweep, not a joint optimization of turbine number and positions.

B. Methods Compared

Six methods are compared, all with the same objective (7), the same bounds $[-r, r]^{2N}$ and the same budget:

- LX-SSA [30]: Algorithm 2 with $\phi = 0$, $\chi = 1$, population 30 and 100 iterations.
- SSA [26]: the same leader/follower dynamics without the Laplace candidate; population 30 and 200 iterations.
- PSO: inertia weight $w = 0.7$ and acceleration coefficients $c_1 = c_2 = 2$, with unbounded velocities and positions clipped to the bounds; population 30 and 200 iterations.
- DE: DE/rand/1/bin with $F = 0.5$ and $CR = 0.9$; population 30 and 200 iterations.
- VNS [22], [40]: basic variable neighbourhood search adapted to continuous coordinates. It starts from the best member of the initial population. Shaking moves $k \in \{1, 2, 3\}$ randomly chosen turbines by Gaussian steps with standard deviation $0.05r$, $0.15r$ and $0.40r$. Each shake is followed by a ten-call first-improvement local search of single-turbine Gaussian moves with an adaptive

TABLE I: Wind Data Sets I and II: Direction Intervals, Weibull Parameters, and Blowing Probabilities

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	13	0.000
2	15	30	2	13	0.010
3	30	45	2	13	0.010
4	45	60	2	13	0.010
5	60	75	2	13	0.010
6	75	90	2	13	0.200
7	90	105	2	13	0.600
8	105	120	2	13	0.010
9	120	135	2	13	0.010
10	135	150	2	13	0.010
11	150	165	2	13	0.010
12	165	180	2	13	0.010
13	180	195	2	13	0.010
14	195	210	2	13	0.010
15	210	225	2	13	0.010
16	225	240	2	13	0.010
17	240	255	2	13	0.010
18	255	270	2	13	0.010
19	270	285	2	13	0.010
20	285	300	2	13	0.010
21	300	315	2	13	0.010
22	315	330	2	13	0.010
23	330	345	2	13	0.010
24	345	360	2	13	0.000

Int.	Start (°)	End (°)	ζ	ψ	Prob.
1	0	15	2	7.0	0.0002
2	15	30	2	5.0	0.0080
3	30	45	2	5.0	0.0227
4	45	60	2	5.0	0.0242
5	60	75	2	5.0	0.0225
6	75	90	2	4.0	0.0339
7	90	105	2	5.0	0.0423
8	105	120	2	6.0	0.0290
9	120	135	2	7.0	0.0617
10	135	150	2	7.0	0.0813
11	150	165	2	8.0	0.0994
12	165	180	2	9.5	0.1394
13	180	195	2	10.0	0.1839
14	195	210	2	8.5	0.1115
15	210	225	2	8.5	0.0765
16	225	240	2	6.5	0.0080
17	240	255	2	4.6	0.0051
18	255	270	2	2.6	0.0019
19	270	285	2	8.0	0.0012
20	285	300	2	5.0	0.0010
21	300	315	2	6.4	0.0017
22	315	330	2	5.2	0.0031
23	330	345	2	4.5	0.0097
24	345	360	2	3.9	0.0317

step (enlarged by 1.5 after a success, reduced by 0.8 after a failure). An improvement is accepted and resets $k = 1$; otherwise k increases cyclically.

- *Multistart SLSQP (MS-SLSQP)*, a gradient-based method: SciPy's sequential least-squares quadratic programming solver [41] minimizes the wake loss with the spacing and boundary constraints imposed explicitly and forward-difference gradients (1-m step). It starts from the members of the initial population in order of F_p and restarts until the budget is exhausted.

SSA, LX-SSA, PSO and DE use the original implementation; VNS and MS-SLSQP were implemented for this study.

C. Budget, Seeds and Pairing

Every run has a budget of exactly 6,030 objective-function calls, counted by a wrapper around the objective. The count includes the initial population and, for MS-SLSQP, the calls used for finite-difference gradients. For LX-SSA this corresponds to 100 iterations, because each follower evaluates two candidates before the population is re-evaluated ($30 + 100 \times 60$ calls); SSA, PSO and DE reach the same count in 200 iterations ($30 + 200 \times 30$). Each method is run with seeds 1–30. Every optimizer seeds the random-number generator and draws its initial population first, so runs with the same seed start from the *same* initial population for all six methods. The runs of a test case are therefore paired by seed.

D. Performance Measures and Statistical Analysis

For each run we record

- the final benchmark objective and wake loss (as a percentage of the wake-free objective);
- whether the final layout is feasible (spacing and boundary satisfied within 10^{-6} m);

- the convergence curve, i.e., the best feasible objective found after every 30 calls;
- the wall-clock time.

Means and standard deviations are computed over the feasible runs.

Within each test case, the 30 seed-paired runs are compared by a run-level Friedman test and by two-sided Wilcoxon signed-rank tests of LX-SSA against each of the other five methods. The five p values of each case are adjusted by Holm's procedure [46], and effect sizes are reported as matched-pairs rank-biserial correlations. For ranking, a feasible run beats an infeasible one, and infeasible runs are compared by their spacing shortfall. Across the 68 cases, the methods are ranked within each case by their mean feasible objective and compared by a case-level Friedman test with the Iman–Davenport correction and Holm-adjusted average-rank post hoc tests, following Demšar [45].

E. Implementation Verification

The objective function was vectorized and checked against the original implementation; the relative difference is below 10^{-6} on random feasible and infeasible layouts. Its wake and energy terms also reproduce the objective values of 720 archived final layouts from an earlier run of the original code (maximum absolute deviation 8.7×10^{-11}). Re-running the original SSA, LX-SSA, PSO and DE code at the archived settings reproduces the archived distributions: 22 of 24 Mann–Whitney comparisons give $p \geq 0.05$, and the means differ by at most 0.21%. This also confirms the Laplace parameters $\phi = 0$ and $\chi = 1$.

F. Computing Environment

All runs were executed on a Linux container with an Intel Xeon processor at 2.80 GHz and 4 virtual cores, four

TABLE II: Case-level Friedman analysis over all 68 benchmark cases (blocks: cases; ranks of the mean feasible objective, 1 = best). Friedman $\chi_F^2 = 135.3$ (5 d.f.), $p = 1.82 \times 10^{-27}$; Iman–Davenport $F_F = 44.3$. Holm-adjusted p values of the average-rank z test against the best-ranked method (VNS) and against LX-SSA; “Best”: cases in which the method alone has the highest mean (seven cases are ties); “Feas.”: percentage of feasible runs.

Method	Avg. rank	Best	Feas. (%)	p vs. VNS	p vs. LX-SSA
VNS	1.60	44	99.6	–	7.95×10^{-13}
SSA	3.15	0	97.6	1.49×10^{-6}	0.031
PSO	3.33	9	86.0	1.44×10^{-7}	0.092
MS-SLSQP	3.87	6	99.9	5.05×10^{-12}	0.748
LX-SSA	3.97	0	96.9	6.36×10^{-13}	–
DE	5.08	2	71.4	1.11×10^{-26}	0.002

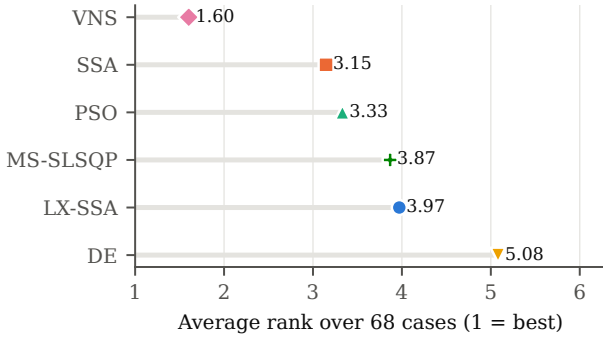


Fig. 4: Average rank of the six methods over the 68 benchmark cases (ranks of the mean feasible objective within each case; 1 = best).

runs in parallel, using Python 3.11.15, NumPy 2.4.6 and SciPy 1.17.1. Each run was timed individually. Scripts, per-run results (including final coordinates and convergence curves) and the figure-generation code accompany the revision.

VII. NUMERICAL RESULTS

This section reports the results of 12,240 runs: 68 test cases, six methods and 30 seed-paired runs, all at 6,030 objective calls. It closes with the Horns Rev 1 site case. Detailed per-case results are given in Tables IV–IX.

A. Overall Comparison

The case-level Friedman test over the 68 test cases rejects the hypothesis that the six methods perform equally ($\chi_F^2 = 135.3$, 5 d.f., $p = 1.8 \times 10^{-27}$; Iman–Davenport $F_F = 44.3$). VNS is clearly the strongest method (Table II, Fig. 4). It has an average rank of 1.60 and alone attains the highest mean objective in 44 of the 68 cases. It is followed by SSA (3.15), PSO (3.33), MS-SLSQP (3.87), LX-SSA (3.97) and DE (5.08). The Holm-adjusted post hoc tests find VNS significantly better than every other method ($p_{\text{Holm}} \leq 1.5 \times 10^{-6}$). The average rank of LX-SSA is significantly worse than that of SSA ($p_{\text{Holm}} = 0.031$) and significantly better than that of DE ($p_{\text{Holm}} = 0.002$); its differences from PSO and MS-SLSQP

TABLE III: Pairwise outcome of LX-SSA against each method over all cases: number of cases in which LX-SSA is significantly better / not significantly different / significantly worse (two-sided Wilcoxon signed-rank test on 30 seed-paired runs, Holm-adjusted over the five comparisons of each case, $\alpha = 0.05$).

Data set	r (m)	Cases	SSA	PSO	DE	VNS	MS-SLSQP
Data Set I	500	9	0/9/0	2/2/5	4/4/1	0/5/4	2/3/4
Data Set I	750	11	0/10/1	3/7/1	8/2/1	0/4/7	3/6/2
Data Set I	1000	14	0/12/2	5/9/0	11/3/0	0/4/10	8/3/3
Data Set II	500	9	0/7/2	2/4/3	4/4/1	0/5/4	1/4/4
Data Set II	750	11	0/8/3	3/7/1	8/2/1	0/5/6	1/7/3
Data Set II	1000	14	0/13/1	3/10/1	11/3/0	0/4/10	3/8/3
All		68	0/59/9	18/39/11	46/18/4	0/27/41	18/31/19

are not significant. LX-SSA never alone attains the highest mean.

The run-level pairwise tests (Table III) give a consistent picture:

- *LX-SSA vs. SSA (mechanism-level ablation)*. LX-SSA is never significantly better. It is statistically indistinguishable in 59 cases and significantly worse in nine (r_{rb} between -0.48 and -0.65), which are mostly dense cases: Data Set I at 750 m/12 and 1000 m/14–15, and Data Set II at 500 m/7 and 9, 750 m/7, 11 and 12, and 1000 m/14. At equal cost, the Laplace follower candidate therefore brings no benefit over the SSA follower update on this benchmark. The likely reason is that it spends two objective calls per follower and iteration, which halves the number of iterations.
- *LX-SSA vs. VNS*. VNS is significantly better in 41 cases, with large effects (median $r_{rb} = -0.87$). The 27 cases without a significant difference all have $N \leq 7$ and are mostly the small cases, in which all methods come close to the wake-free optimum.
- *LX-SSA vs. PSO* (18 better, 39 not different, 11 worse). LX-SSA wins mainly in the dense cases, where PSO often fails to find a feasible layout. It loses in small and medium cases ($N = 3$ –7), where PSO converges to better layouts.
- *LX-SSA vs. MS-SLSQP* (18/31/19). LX-SSA is better at intermediate turbine counts, mostly for the 1000-m farm, and MS-SLSQP is better in the densest cases ($N \geq 7$ at 500 m, $N \geq 10$ at 750 m and $N \geq 13$ at 1000 m).
- *LX-SSA vs. DE* (46/18/4). LX-SSA is better in most cases; the four losses are all three-turbine cases.

B. Solution Quality by Farm Size and Turbine Count

Figure 5 and Tables IV–IX show how the solution quality depends on the farm radius and the turbine count. For $N \leq 3$ all methods stay within 0.3% of the wake-free objective. The wake loss then grows with N and falls with the farm radius, and it is higher for Data Set II, whose wind comes from a broader range of directions, than for the strongly directional Data Set I. The differences between the methods grow with N . In the largest cases the lowest mean wake loss is obtained by VNS or MS-SLSQP; for example:

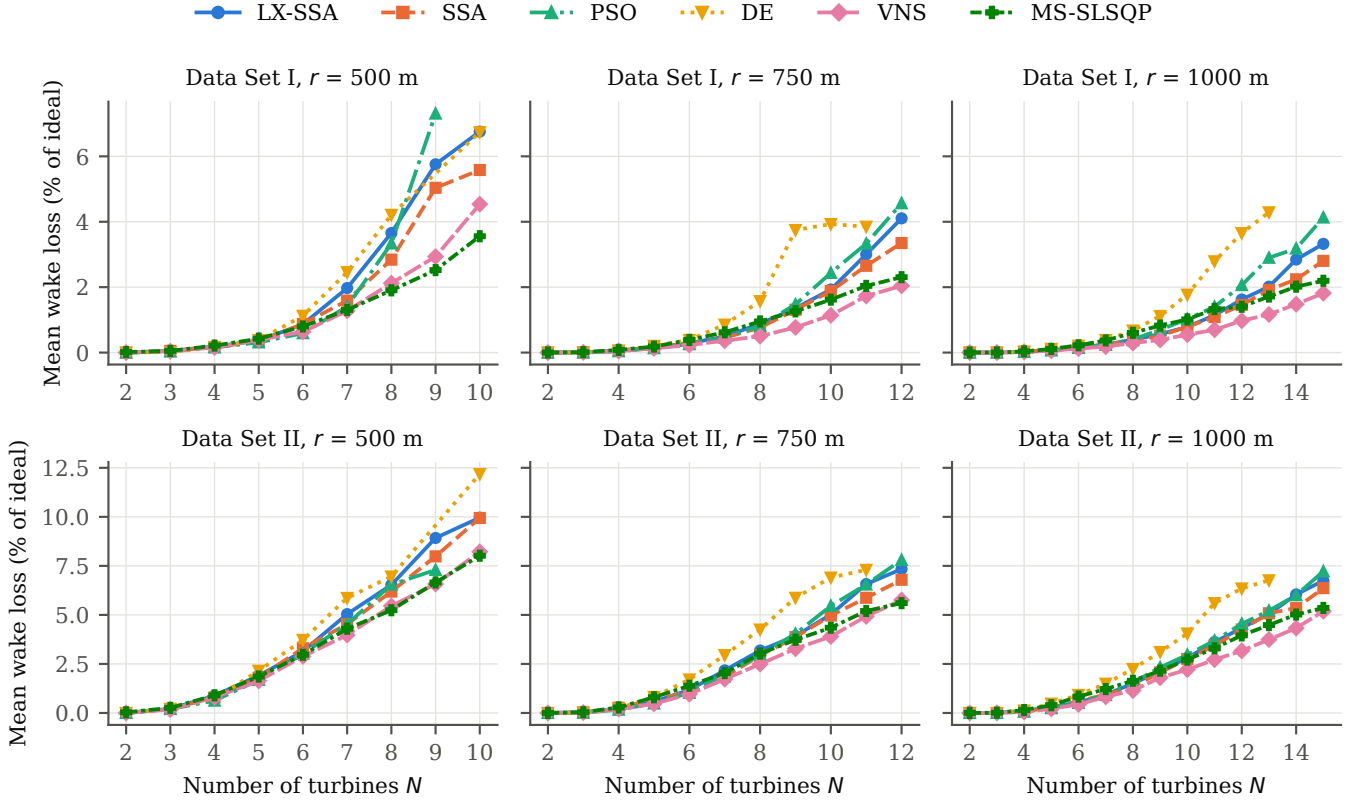


Fig. 5: Mean wake loss (percentage of the wake-free objective) of the feasible runs versus the prescribed number of turbines; a point is omitted when a method has no feasible run.

- Data Set I, 1000 m/15: VNS 1.82%, MS-SLSQP 2.20%, SSA 2.80%, LX-SSA 3.32%;
- Data Set I, 500 m/10: MS-SLSQP 3.56%, VNS 4.54%, SSA 5.58%, LX-SSA 6.76%;
- Data Set II, 1000 m/15: VNS 5.19%, MS-SLSQP 5.36%, SSA 6.36%, LX-SSA 6.75%.

In every largest case, LX-SSA has a higher mean wake loss than SSA.

C. Feasibility

Figure 6 reports how often each method ends with a feasible layout. Up to $N = 6$ (500 m), $N = 8$ (750 m) and $N = 10$ (1000 m) all methods are always feasible. Beyond these counts the methods separate clearly:

- *DE* and *PSO* lose feasibility first. *DE* finds no feasible layout at 500 m with $N = 9$, at 750 m with $N = 12$ or at 1000 m with $N \geq 14$. *PSO* succeeds in 2 of 30 runs at 500 m/9 and in none at 500 m/10.
- *LX-SSA* and *SSA* remain feasible much longer. They drop to 13 and 12 feasible runs at 500 m/10 and 21 and 25 at 500 m/9, but stay at 28–30 in all 750-m and 1000-m cases.
- *VNS* and *MS-SLSQP* are the most reliable, with at least 27 feasible runs in every case.

For LX-SSA, SSA, PSO, DE and VNS the feasibility counts are identical for the two wind data sets. The constraints do

not depend on the wind data, and while every candidate is infeasible the penalty term dominates F_p , so the search follows the same path until a feasible layout is found. MS-SLSQP, which uses the wind-dependent wake loss from the start, differs slightly between the data sets.

D. Convergence Behaviour

Figures 7 and 8 show the median best feasible wake loss as a function of the number of objective calls.

- *LX-SSA* and *SSA* converge early and then stagnate. In the six moderate cases they reach within 0.1% of their final objective after a median of 2,900–3,800 calls, so the second half of the budget brings little improvement.
- *PSO* and *DE* converge more slowly: *PSO* needs 4,300–5,100 calls and *DE* 3,600–5,600.
- *VNS* needs 2,900–4,600 calls and still improves late in the budget, which is consistent with its local-search refinement.
- *MS-SLSQP* finds feasible layouts earliest. In the largest cases its median first feasible layout appears after 150–690 calls, compared with 1,500–2,900 for VNS and 2,800–3,700 for LX-SSA and SSA.

In the largest cases the median LX-SSA curve lies above the SSA curve for most of the budget.

TABLE IV: Data Set I, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28087.2 (4.6)
3	42137.2	42114.8 (14.2)	42114.9 (10.3)	42124.7 (8.5)	42130.7 (8.8)	42119.7 (11.8)	42116.0 (10.1)
4	56182.9	56078.8 (22.6)	56086.8 (19.8)	56094.3 (13.3)	56092.8 (29.8)	56094.2 (28.5)	56063.2 (30.9)
5	70228.7	69972.6 (72.0)	69994.0 (34.0)	70010.5 (27.2)	69960.5 (78.6)	69955.1 (126.5)	69930.5 (113.0)
6	84274.4	83544.9 (465.3)	83535.6 (372.5)	83775.8 (154.1)	83326.7 (761.5)	83738.2 (213.4)	83601.4 (213.1)
7	98320.2	96380.6 (1013.8)	96757.0 (692.4)	96935.0 (485.2)	95903.3 (1316.3) ¹⁵	97066.5 (484.7)	97037.0 (484.3)
8	112365.9	108258.3 (1704.3) ²⁸	109172.6 (1362.4)	108616.6 (1480.8) ²⁴	107632.7 (1592.8) ²	109981.7 (835.2)	110226.6 (585.5)
9	126411.6	119134.2 (2666.4) ²¹	120043.1 (2755.8) ²⁵	117175.2 (7148.7) ²	– ⁰	122702.8 (1513.8) ²⁹	123222.8 (812.6)²⁸
10	140457.4	130967.4 (3191.0) ¹³	132616.3 (2328.0) ¹²	– ⁰	130994.8 (0.0) ¹	134084.5 (1603.0) ²⁷	135461.0 (1291.9)

TABLE V: Data Set I, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.0 (1.6)
3	42137.2	42134.2 (4.8)	42134.4 (4.1)	42137.0 (1.1)	42136.8 (1.5)	42134.5 (4.9)	42132.3 (5.0)
4	56182.9	56149.9 (14.0)	56153.8 (17.9)	56160.0 (14.2)	56146.9 (17.5)	56156.3 (14.4)	56137.1 (16.4)
5	70228.7	70118.1 (28.6)	70122.2 (26.3)	70135.0 (22.6)	70083.9 (31.7)	70137.4 (36.3)	70100.5 (42.3)
6	84274.4	84033.5 (58.3)	84035.2 (58.4)	84059.1 (42.0)	83948.2 (74.1)	84079.8 (64.2)	83947.6 (95.9)
7	98320.2	97840.0 (167.7)	97875.2 (169.3)	97872.3 (131.8)	97479.1 (285.1)	97963.6 (80.6)	97710.2 (204.8)
8	112365.9	111374.0 (460.8)	111506.5 (334.5)	111448.4 (551.6)	110595.8 (445.6)	111801.5 (162.3)	111296.2 (217.8)
9	126411.6	124698.0 (960.4)	124756.2 (610.5)	124550.0 (794.2)	121674.9 (2276.5) ²³	125437.8 (379.2)	124818.4 (466.3)
10	140457.4	137748.6 (1116.3)	137817.6 (1160.1)	137036.8 (1051.3) ²⁸	134937.7 (998.5) ⁴	138856.6 (431.7)	138184.7 (595.3)
11	154503.1	149863.8 (1886.6) ²⁹	150404.1 (1042.2) ²⁹	149356.6 (1849.0) ²⁰	148555.7 (1134.8) ³	151831.7 (908.0)	151352.9 (560.2)
12	168548.8	161630.5 (1988.2) ²⁸	162903.5 (1685.8)	160854.7 (2108.4) ⁷	– ⁰	165106.7 (762.7)	164663.8 (674.6)

TABLE VI: Data Set I, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	28091.5	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.5 (0.0)	28091.3 (0.7)
3	42137.2	42136.7 (1.4)	42136.7 (1.6)	42137.2 (0.0)	42137.2 (0.0)	42137.0 (0.9)	42135.9 (2.6)
4	56182.9	56172.9 (7.8)	56169.4 (9.2)	56173.0 (6.8)	56166.9 (8.4)	56170.1 (13.3)	56162.5 (10.5)
5	70228.7	70183.3 (21.0)	70174.0 (19.7)	70182.6 (13.0)	70148.7 (22.4)	70186.7 (23.4)	70151.4 (32.7)
6	84274.4	84142.7 (23.5)	84149.9 (29.7)	84152.3 (22.2)	84096.1 (24.0)	84171.1 (30.2)	84087.4 (92.6)
7	98320.2	98078.4 (82.1)	98084.7 (47.0)	98083.0 (35.8)	97942.3 (67.9)	98141.0 (51.2)	97955.8 (136.6)
8	112365.9	111924.6 (164.7)	111947.1 (125.7)	111915.5 (125.3)	111608.3 (168.9)	112038.8 (78.6)	111688.0 (181.9)
9	126411.6	125701.5 (265.4)	125761.1 (168.4)	125531.1 (324.5)	124996.3 (282.9)	125923.2 (134.4)	125366.6 (282.7)
10	140457.4	139354.2 (388.5)	139348.0 (433.0)	139018.6 (478.6)	137969.0 (623.0)	139687.2 (221.5)	139031.6 (341.6)
11	154503.1	152736.8 (558.8)	152825.1 (385.8)	152337.8 (768.8) ²⁹	150189.7 (1068.9) ²⁹	153426.4 (348.5)	152445.9 (402.0)
12	168548.8	165814.4 (827.9)	166091.9 (816.6)	165072.5 (1367.5) ²⁸	162390.4 (1765.9) ¹⁷	166906.5 (459.2)	166188.4 (681.0)
13	182594.6	178913.6 (967.4)	179101.5 (1028.9)	177302.0 (2352.6) ²²	174763.4 (1389.6) ⁴	180466.7 (599.8)	179465.2 (506.2)
14	196640.3	191046.6 (1754.6)	192235.7 (1504.8)	190385.1 (1544.4) ¹⁸	– ⁰	193732.7 (887.0)	192677.8 (595.4)
15	210686.1	203681.8 (1899.5) ²⁹	204777.6 (1420.5)	201985.5 (1678.3) ⁹	– ⁰	206859.5 (910.3)	206059.6 (761.8)

TABLE VII: Data Set II, 500-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “–” means no feasible run.

<i>N</i>	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14626.0 (5.6)
3	21946.1	21899.7 (9.5)	21900.9 (11.1)	21901.5 (6.3)	21901.2 (5.5)	21906.5 (23.0)	21888.8 (45.5)
4	29261.5	29004.2 (87.4)	29028.7 (93.8)	29077.9 (64.5)	29044.2 (61.5)	29028.5 (66.7)	28998.9 (104.2)
5	36576.9	35880.1 (168.6)	35918.9 (74.0)	35959.4 (119.0)	35789.6 (193.8)	35976.5 (173.2)	35896.9 (110.0)
6	43892.3	42512.1 (227.9)	42444.3 (224.5)	42568.5 (159.7)	42257.5 (374.9)	42626.6 (192.4)	42590.7 (236.0)
7	51207.6	48626.5 (340.5)	48887.9 (304.1)	48885.9 (278.3)	48203.9 (767.8) ¹⁵	49171.8 (236.9)	49007.9 (208.2)
8	58523.0	54699.5 (620.7) ²⁸	54900.5 (471.3)	54693.1 (404.3) ²⁴	54451.3 (618.7) ²	55331.0 (323.5)	55457.2 (287.3)
9	65838.4	59963.3 (652.2) ²¹	60581.0 (844.8) ²⁵	61035.6 (137.9) ²	– ⁰	61509.3 (538.6)²⁹	61457.0 (370.7)
10	73153.8	65868.7 (586.0) ¹³	65880.7 (761.6) ¹²	– ⁰	64240.2 (0.0) ¹	67141.9 (581.4) ²⁷	67288.3 (300.1)²⁹

E. Distribution of Results

Figure 9 shows the distribution of the final wake loss for the moderate cases. VNS has the lowest median in five of

the six cases (PSO has it in Data Set I at 500 m/6), and DE has the highest median in all six. The LX-SSA and SSA distributions overlap strongly: LX-SSA has the lower median

TABLE VIII: Data Set II, 750-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “—” means no feasible run.

N	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.4 (1.5)
3	21946.1	21937.3 (8.1)	21940.0 (8.2)	21943.2 (5.4)	21942.5 (5.8)	21941.5 (6.8)	21938.2 (12.2)
4	29261.5	29201.4 (25.7)	29182.5 (36.2)	29213.0 (20.4)	29195.3 (26.2)	29202.4 (32.5)	29179.0 (59.2)
5	36576.9	36366.8 (74.3)	36332.0 (91.4)	36400.0 (62.2)	36277.7 (105.9)	36405.5 (82.8)	36283.6 (98.0)
6	43892.3	43377.4 (175.7)	43419.5 (153.1)	43441.8 (107.5)	43142.2 (144.8)	43470.4 (120.0)	43283.1 (153.3)
7	51207.6	50095.4 (193.6)	50264.8 (202.5)	50188.2 (199.5)	49696.6 (156.3)	50324.3 (194.4)	50164.7 (205.9)
8	58523.0	56659.2 (332.0)	56806.2 (303.6)	56786.4 (311.8)	56033.1 (331.4)	57063.0 (286.7)	56752.4 (256.7)
9	65838.4	63252.7 (342.8)	63291.6 (402.5)	63180.2 (411.7)	61970.8 (763.0) ²³	63680.4 (415.5)	63381.6 (322.9)
10	73153.8	69480.7 (385.7)	69520.3 (512.6)	69159.6 (436.1) ²⁸	68103.2 (423.7) ⁴	70300.6 (409.0)	69972.9 (458.1)
11	80469.2	75181.6 (681.1) ²⁹	75742.5 (677.1) ²⁹	75199.6 (585.7) ²⁰	74596.6 (476.3) ³	76504.5 (395.7)	76284.4 (433.9)
12	87784.5	81341.8 (792.1) ²⁸	81822.1 (625.7)	80936.3 (798.4) ⁷	— ⁰	82731.2 (470.9)	82867.2 (399.2)

TABLE IX: Data Set II, 1000-m farm: benchmark objective, mean (SD) over the feasible runs of 30 seed-paired runs at 6,030 objective calls. Bold: highest mean. A superscript gives the number of feasible runs when fewer than 30; “—” means no feasible run.

N	Ideal	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP
2	14630.8	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.8 (0.0)	14630.5 (1.0)
3	21946.1	21945.3 (2.8)	21945.6 (1.8)	21945.8 (2.0)	21946.1 (0.0)	21945.6 (1.7)	21943.5 (4.8)
4	29261.5	29228.0 (14.9)	29232.5 (18.5)	29243.3 (13.6)	29225.9 (12.1)	29246.6 (13.5)	29218.7 (24.2)
5	36576.9	36468.2 (46.1)	36485.9 (37.5)	36486.0 (36.0)	36402.8 (39.4)	36499.6 (38.8)	36432.0 (50.4)
6	43892.3	43656.6 (73.3)	43626.4 (74.5)	43665.8 (65.6)	43488.5 (69.7)	43700.2 (82.3)	43524.8 (86.8)
7	51207.6	50727.3 (119.3)	50731.8 (118.8)	50739.2 (134.6)	50442.7 (136.0)	50785.0 (110.8)	50582.5 (165.5)
8	58523.0	57645.1 (186.6)	57739.3 (142.0)	57616.2 (157.7)	57210.0 (191.8)	57854.1 (163.9)	57560.8 (206.6)
9	65838.4	64428.7 (273.1)	64476.4 (203.0)	64308.4 (234.8)	63790.1 (227.2)	64652.4 (279.3)	64419.1 (268.8)
10	73153.8	71099.4 (266.9)	71139.4 (336.8)	70985.2 (314.4)	70190.1 (286.9)	71535.7 (321.0)	71161.4 (275.8)
11	80469.2	77561.6 (498.0)	77697.5 (425.8)	77501.2 (352.1) ²⁹	75960.9 (577.6) ²⁹	78281.9 (285.9)	77812.4 (323.7)
12	87784.5	83981.2 (515.0)	84049.6 (660.3)	83795.7 (565.1) ²⁸	82211.6 (905.2) ¹⁷	85002.5 (322.8)	84312.7 (466.3)
13	95099.9	90303.5 (674.2)	90267.0 (482.6)	90139.1 (533.9) ²²	88665.1 (496.0) ⁴	91538.1 (357.6)	90837.9 (572.0)
14	102415.3	96229.6 (1019.3)	96936.5 (669.3)	96266.2 (730.7) ¹⁸	— ⁰	97983.6 (409.3)	97276.4 (405.3)
15	109730.7	102327.4 (838.7) ²⁹	102746.3 (827.8)	101804.8 (571.6) ⁹	— ⁰	104036.5 (619.7)	103853.3 (475.2)

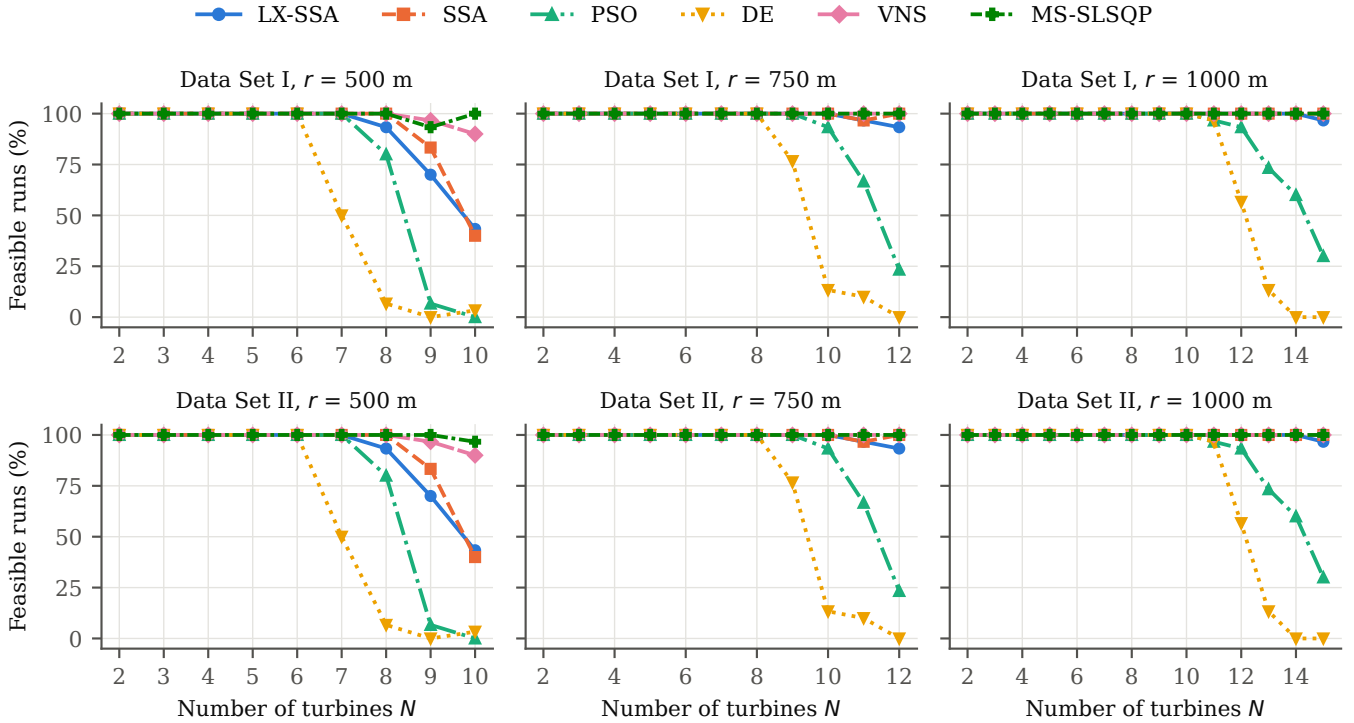


Fig. 6: Percentage of the 30 runs that end with a feasible layout (all turbines inside the farm and at least 308 m apart).

in three cases and SSA in the other three, in line with the mostly non-significant paired tests between them.

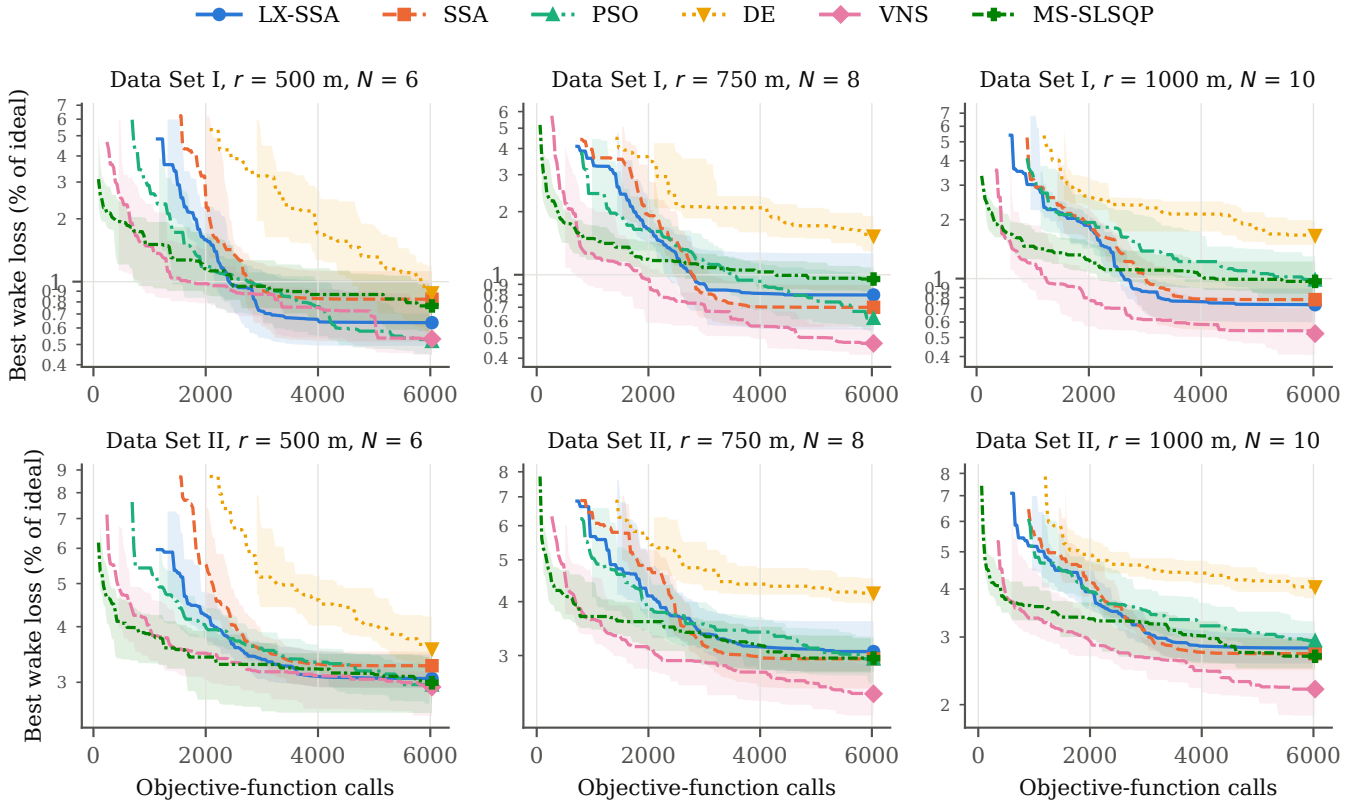


Fig. 7: Convergence for moderate turbine counts: median (line) and interquartile range (band) over the 30 runs of the best feasible wake loss found so far, versus the number of objective calls (logarithmic scale). A curve starts once at least half of the runs have found a feasible layout.

F. Optimized Layouts

Figure 10 compares, for the largest turbine count of each farm, the best LX-SSA layout with the best layout found by any method; the best overall layout comes from VNS or MS-SLSQP in all six cases. Most turbines are placed close to the boundary, which maximizes their mutual distances, while the remaining turbines occupy the interior at positions staggered with respect to the dominant wind directions.

G. Computational Cost

At an equal number of objective calls the methods need almost the same time: 0.40 ms per call for LX-SSA and SSA, 0.41 ms for VNS and PSO, 0.47 ms for DE and 0.49 ms for MS-SLSQP, whose quadratic subproblems add a small overhead. A 6,030-call run of LX-SSA takes 1.7–3.6 s on average, depending on N ; the slowest single run of any method (MS-SLSQP) took 6.3 s. The objective evaluation dominates the cost, so the Laplace step of LX-SSA adds no measurable overhead. The complete study of 12,240 runs took 2.2 h on four cores.

H. Horns Rev 1 Site Case

To bring the validation closer to a real project, the Horns Rev 1 offshore farm was modeled with its real turbine,

wind climate and site outline, all taken from the PyWake distribution [42]:

- the installed layout of 80 Vestas V80 turbines (rotor diameter 80 m, 2 MW) and the V80 power and thrust-coefficient curves;
- the measured 12-sector wind climate (Weibull parameters and frequency per sector);
- the parallelogram spanned by the installed turbines as the site outline.

The farm model keeps the Jensen top-hat wake (with $k = 0.04$, a value commonly used offshore) and root-sum-square superposition. It uses the tabulated power curve with a 25 m/s cut-out, a speed-dependent thrust coefficient evaluated at the free-stream speed, 5° direction bins and 1 m/s speed bins. As an independent check, this model gives the installed layout an AEP of 664.6 GWh/yr with a wake loss of 10.95%, compared with 662.5 GWh/yr and 10.96% from PyWake’s Jensen model with the same settings, a difference below 0.3%. Two problems were solved with the same six methods, 6,030 calls per run and the 4D (320 m) minimum spacing: the 16-turbine block that PyWake uses as an example (30 seeds) and the complete 80-turbine farm (10 seeds, because one run takes about 100 s).

Table X and Figs. 11 and 12 lead to three observations.

- *No method improves on the installed layout within the budget.* In the 16-turbine block VNS reaches

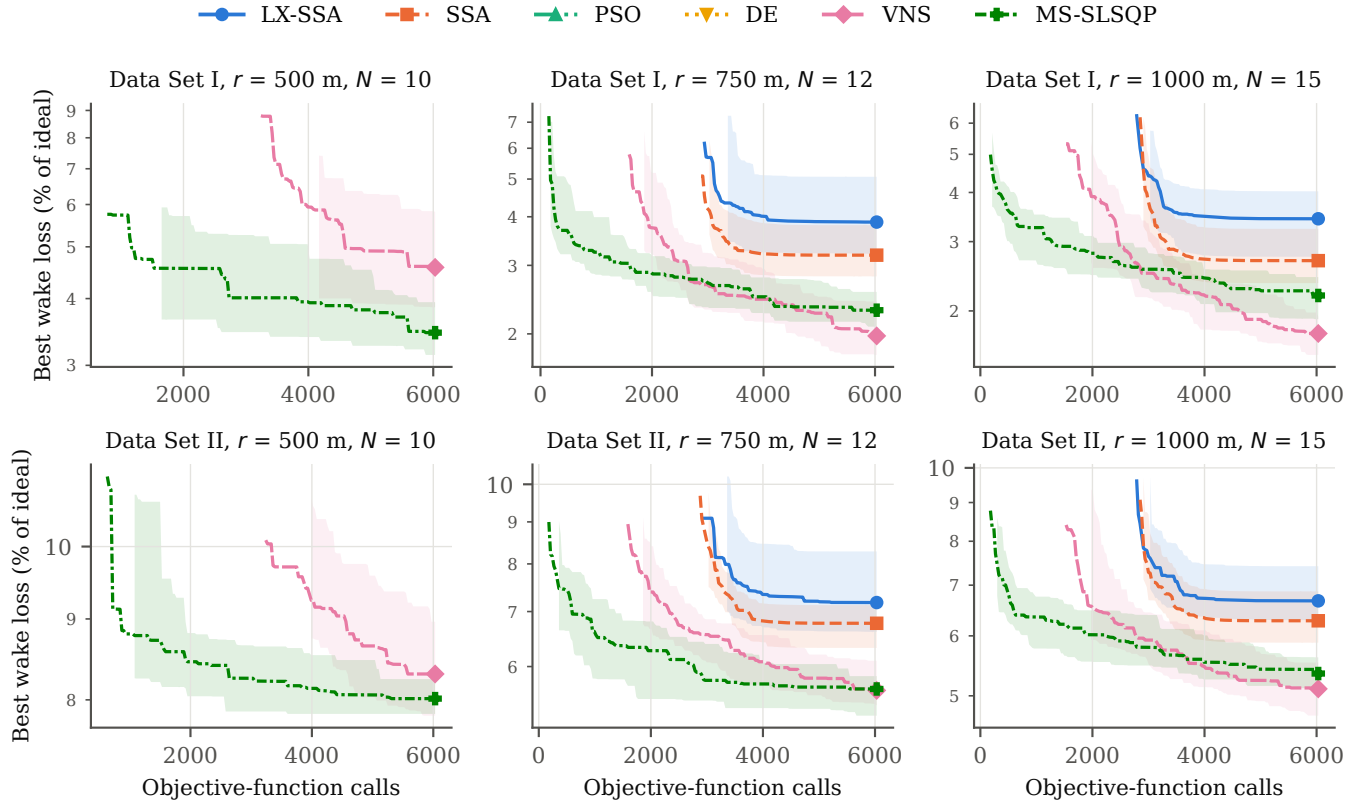


Fig. 8: Convergence for the largest turbine count of each farm (as Fig. 7). Methods for which fewer than half of the runs find a feasible layout within the budget have no curve (PSO and DE, and LX-SSA and SSA at 500 m/10).

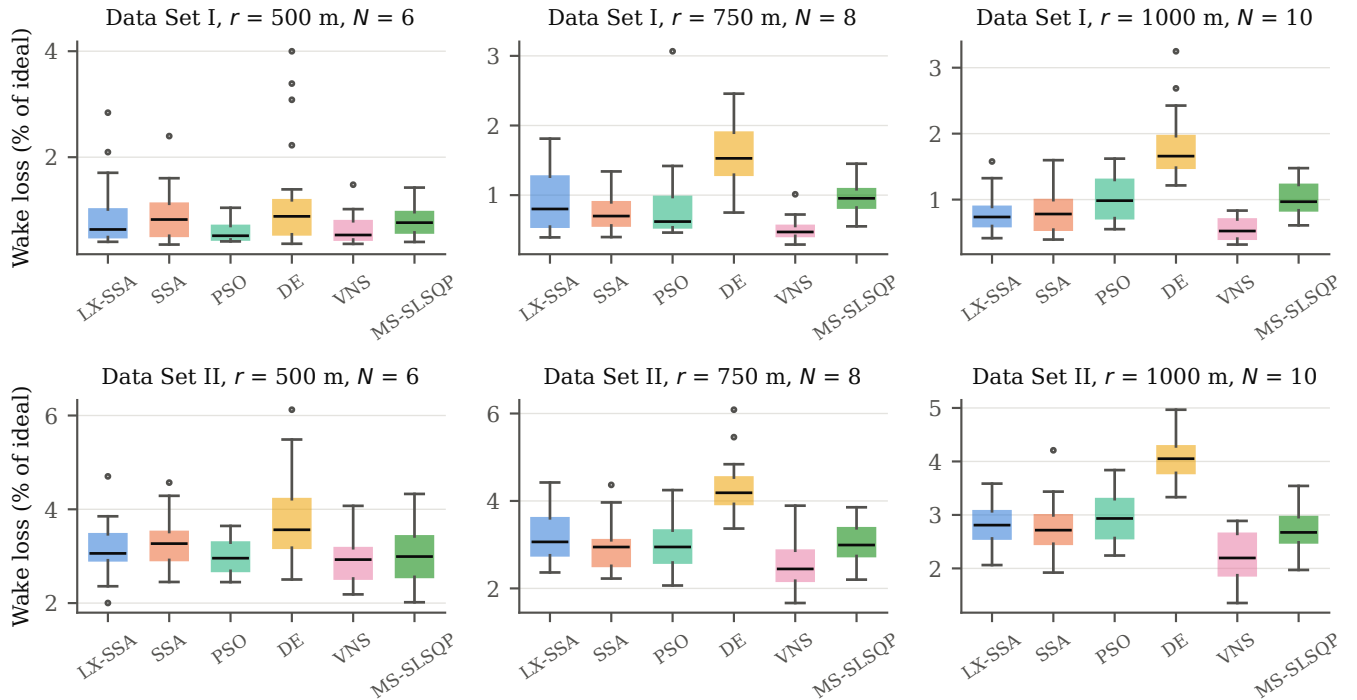


Fig. 9: Distribution of the final wake loss of the feasible runs for moderate turbine counts (30 runs per method).

137.97 GWh/yr on average and LX-SSA 136.46 GWh/yr,

compared with 139.51 GWh/yr for the installed 7D grid.

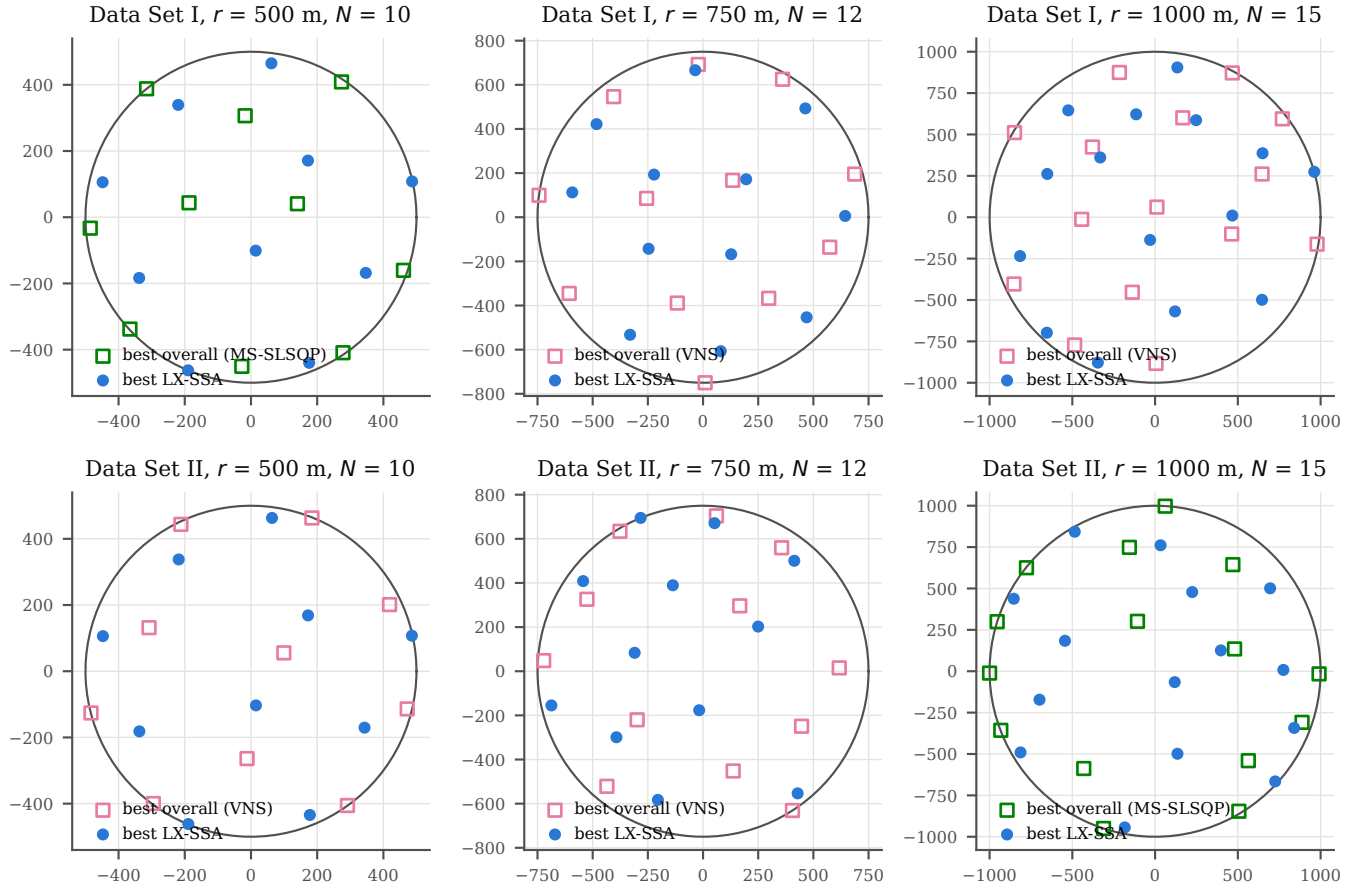


Fig. 10: Best layouts for the largest turbine count of each farm: best LX-SSA run (filled circles) and best run of any method (open squares, method in the legend). Circle: farm boundary. Coordinates are listed in Appendix A.

TABLE X: Horns Rev 1 site case (real Vestas V80 power and thrust curves, measured 12-sector wind climate, installed farm outline; Jensen wake with $k = 0.04$; minimum spacing $4D = 320$ m): AEP in GWh/yr, mean (SD) over the feasible runs at 6,030 objective calls (30 seeds for the 16-turbine block, 10 for the 80-turbine farm), mean wake loss relative to the wake-free AEP (“Loss”, %), feasible runs, and Holm-adjusted Wilcoxon signed-rank p of LX-SSA vs. each method (16-turbine block; run-level Friedman $p = 4.43 \times 10^{-24}$).

Layout / method	16-turbine block (wake-free 149.26 GWh/yr)				80-turbine farm (wake-free 746.3 GWh/yr)		
	AEP	Loss	Feas.	p_{Holm}	AEP	Loss	Feas.
Installed layout	139.51	6.53	–	–	664.6	10.95	–
LX-SSA	136.46 (0.90)	8.58	28/30	–	–	–	0/10
SSA	136.66 (1.00)	8.44	30/30	0.262	–	–	0/10
PSO	–	–	0/30	9.31×10^{-9}	–	–	0/10
DE	–	–	0/30	9.31×10^{-9}	–	–	0/10
VNS	137.97 (0.80)	7.57	29/30	1.68×10^{-4}	–	–	0/10
MS-SLSQP	136.80 (0.77)	8.35	30/30	0.085	653.9 (2.6)	12.39	10/10

- The algorithm ordering matches the benchmark results. VNS is significantly better than LX-SSA ($p_{\text{Holm}} = 1.7 \times 10^{-4}$); LX-SSA does not differ significantly from SSA ($p_{\text{Holm}} = 0.26$) or MS-SLSQP ($p_{\text{Holm}} = 0.085$); and PSO and DE never find a feasible layout inside the parallelogram. The convergence curves show the same behaviour as for the benchmark: VNS improves steadily, whereas LX-SSA and SSA stagnate after about 4,000 calls.
- The full farm exceeds what the penalty-based methods can handle at this budget. With 160 variables and 3,160 spacing constraints, none of the 50 runs of LX-SSA, SSA, PSO, DE and VNS returns a feasible layout. MS-SLSQP, which handles the constraints explicitly, is feasible in all ten runs, but its mean AEP (653.9 GWh/yr) remains 1.6% below that of the installed layout.

The benchmark-scale conclusions therefore do not transfer directly to realistic farm sizes: such farms require larger

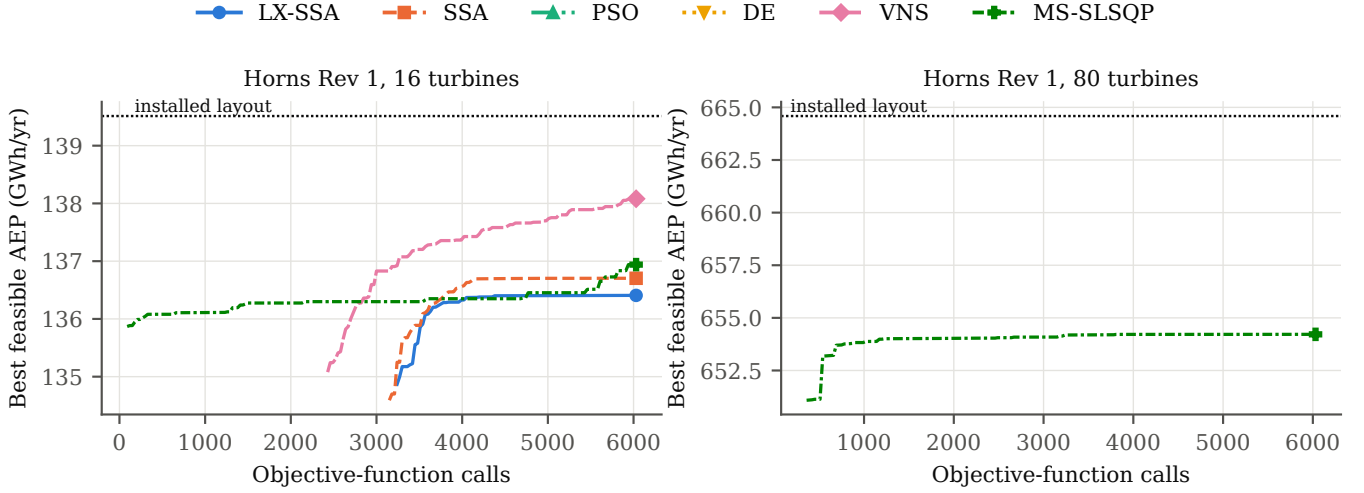


Fig. 11: Horns Rev 1 site case: median best feasible AEP versus objective calls (16-turbine block: 30 runs; 80-turbine farm: 10 runs); dotted line: installed layout. Only MS-SLSQP finds feasible layouts for the 80-turbine farm.

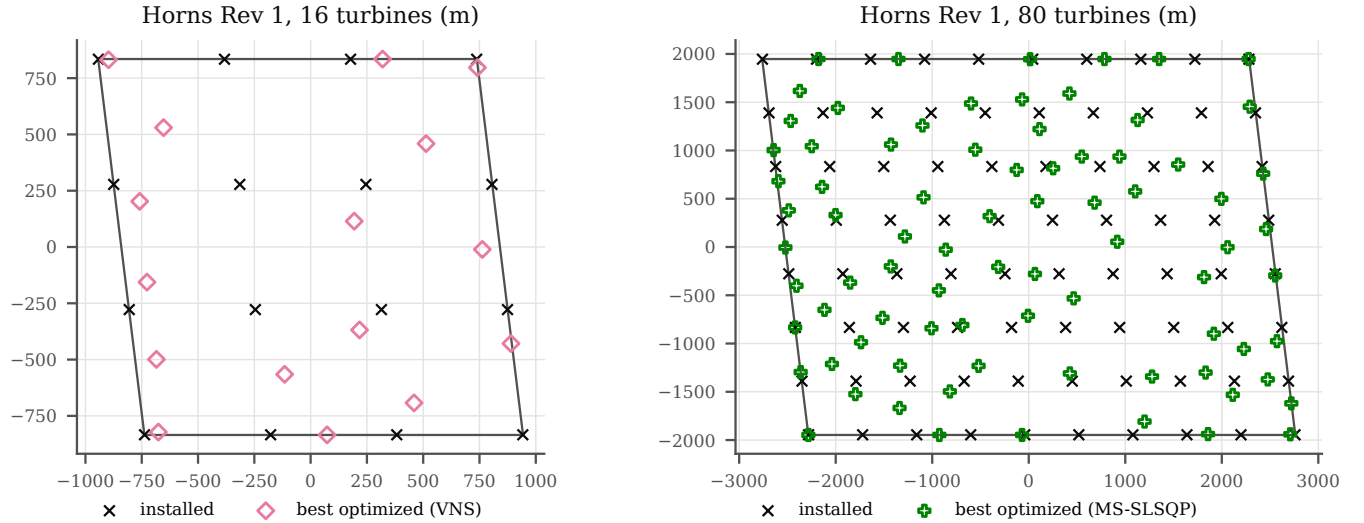


Fig. 12: Horns Rev 1: installed layout (crosses) and best optimized layout (open markers) inside the installed farm outline; local coordinates in metres.

budgets, feasible initialization or explicit constraint handling, and a regular installed layout is a strong reference that none of the tested methods beats within 6,030 calls.

VIII. SENSITIVITY TO THE MODELLING ASSUMPTIONS

A. Power Curve and Wake Model

The benchmark uses a piecewise-linear power curve and Jensen's wake model. To test how far the comparison depends on these choices, every feasible final layout of the study was re-evaluated with three alternative models; the layouts were not re-optimized:

- a cubic power curve [43],

$$f_c(s) = P_{\text{rated}} \frac{s^3 - s_{\text{cut-in}}^3}{s_{\text{rated}}^3 - s_{\text{cut-in}}^3} \quad (s_{\text{cut-in}} \leq s \leq s_{\text{rated}}),$$

with the same cut-in speed, rated speed and rated power;

- the same cubic curve with a 25 m/s cut-out;
- the Gaussian wake model of Bastankhah and Porté-Agel [44],

$$\delta_{ij} = \left(1 - \sqrt{1 - C_T/(8\sigma^2/D^2)}\right) \exp(-r_{\perp}^2/(2\sigma^2)),$$

$$\sigma/D = k^*x/D + 0.2\sqrt{\beta},$$

with $\beta = \frac{1}{2}(1 + \sqrt{1 - C_T})/\sqrt{1 - C_T}$ and $k^* = 0.04$, evaluated at the rotor center with the same superposition and Weibull scaling as the benchmark.

Table XI shows two different kinds of sensitivity.

- *Absolute energy is strongly model-dependent.* The linear benchmark curve overstates expected power relative to the cubic curve by 17.7% for Data Set I and 36.9%

TABLE XI: Robustness of the fresh results to the benchmark model. All feasible final layouts are re-evaluated (not re-optimized) with alternative power-curve and wake models. Δ : mean relative change of the objective (expected power) with respect to the benchmark model; average rank of each method over the 68 cases (methods with at least five feasible runs); $\bar{\tau}$: mean Kendall rank correlation between the benchmark and alternative orderings of the methods within a case; “Same best”: percentage of cases in which the best method is unchanged.

Model	Δ (%)		Average rank							$\bar{\tau}$	Same best (%)
	DS I	DS II	LX-SSA	SSA	PSO	DE	VNS	MS-SLSQP			
Benchmark (linear curve, Jensen)	+0.0	+0.0	3.87	3.50	3.29	4.99	1.59	3.76	–	–	
Cubic power curve	-17.7	-36.9	3.84	3.46	3.24	4.96	1.66	3.84	0.97	96	
Cubic curve with 25 m/s cut-out	-21.5	-37.1	3.84	3.47	3.23	4.98	1.68	3.81	0.97	97	
Gaussian wake ($k^* = 0.04$)	-0.2	-0.3	4.19	3.51	2.82	5.06	2.22	3.19	0.58	49	

TABLE XII: Largest turbine count for which a layout satisfying the circular boundary and the minimum spacing was constructed (multi-start packing search; a constructive lower bound on geometric capacity), compared with the largest tested N .

r (m)	4D (308 m)	5D (385 m)	6D (462 m)	Largest tested N
500	13	8	7	10
750	26	19	13	12
1000	42	29	21	15

for Data Set II (21.5% and 37.1% with the cut-out). The absolute energy values of this paper are therefore benchmark values, not turbine-specific AEP predictions.

- *The algorithm comparison is robust to the power curve but not to the wake model.* Under the cubic curve the ordering of the methods within a case is almost unchanged (mean Kendall $\bar{\tau} = 0.97$; same best method in 96% of the cases). Under the Gaussian wake model the within-case orderings change substantially ($\bar{\tau} = 0.58$; same best method in only 49% of the cases), because layouts optimized for the sharp Jensen wake cone are penalized differently by the smooth Gaussian deficit. Nevertheless VNS keeps the best average rank (2.22), and LX-SSA remains fifth (4.19). Whether the ordering persists when layouts are *optimized* with a higher-fidelity wake model remains open.

B. Minimum Spacing

The 4D spacing is a legacy benchmark value. Its influence was examined in three ways.

Geometric capacity. Table XII reports the largest turbine count for which a multi-start packing search constructed a layout satisfying both the circular boundary and the minimum distance. These are constructive lower bounds, and the 500-m values coincide with the known optimal circle-packing results. Under 4D, all tested turbine counts are well below capacity. At 5D the 500-m farm holds at most 8 turbines and at 6D only 7, so the tested 500-m cases with $N \geq 9$ (at 5D) or $N \geq 8$ (at 6D) would be infeasible. All tested 750-m and 1000-m cases remain geometrically feasible up to 6D.

Final layouts. The optimized layouts are frequently active at the 4D constraint. Of the 11,247 feasible final layouts of the study, 48.7% also satisfy 5D and 35.7% satisfy 6D. For

$N = 6$ –10 these fractions fall to 29.6% and 11.3%, and for $N = 11$ –15 to 3.1% and 0%. Optimized layouts therefore cannot be transferred to a site with larger spacing requirements without re-optimization.

Re-optimization. Six representative cases were re-optimized with LX-SSA and SSA at 5D and 6D (6,030 calls, 30 seed-paired runs). Table XIII shows that larger spacing has little effect in the four-turbine 500-m cases. In the 750-m eight-turbine cases, however, the mean LX-SSA objective falls from 4D to 6D by 0.95% (Data Set I) and 0.22% (Data Set II), and the wake loss rises from 0.88% to 1.85% and from 3.18% to 3.47%. The spacing rule does not reverse the LX-SSA–SSA comparison: the only significant differences (at 6D) favor SSA.

C. Boundary Handling

The implementation clips coordinates to the bounding square and penalizes points outside the circle (Section III-C). An otherwise identical SSA that instead projects every turbine radially onto the circle, and ranks candidates by the feasibility-first rules, attains higher mean objectives than the original SSA in six representative cases at 3,030 calls. The gains range from 26 to 649 objective units (Mann–Whitney $p < 10^{-4}$ in every case). The boundary-handling rule therefore affects results by more than the differences between the salp-swarm algorithms, and it is held fixed for all methods in this study.

IX. COMPUTATIONAL COMPLEXITY

Let N_t be the number of turbines, N_p the population size, T the number of iterations, N_θ the number of wind-direction bins and N_s the number of wind-speed bins. A direct wake evaluation contains the pairwise turbine interactions for every direction and speed bin, so the dominant cost of one objective evaluation is

$$C_{\text{obj}} = \mathcal{O}(N_t^2 N_\theta N_s),$$

and a population-based run costs $\mathcal{O}(TN_p N_t^2 N_\theta N_s)$, or equivalently $\mathcal{O}(B N_t^2 N_\theta N_s)$ for a budget of B objective calls. The boundary repair is $\mathcal{O}(N_t)$ per candidate and the spacing check $\mathcal{O}(N_t^2)$; both are dominated by the wake evaluation. The Laplace draw and candidate update of LX-SSA are linear in the decision dimension. LX-SSA does, however, spend two objective calls per follower and iteration, so at a fixed budget it performs half as many iterations as SSA. The measured costs in Section VII-G confirm that time per call is almost identical across methods and grows with N_t .

TABLE XIII: Minimum-spacing sensitivity with the authors' LX-SSA and SSA code (6,030 calls, 30 seed-paired runs per cell): mean benchmark objective, Wilcoxon signed-rank p for LX-SSA vs. SSA, and the mean LX-SSA wake loss as a percentage of the ideal objective at $4D$ and $6D$. A superscript gives the number of feasible runs when fewer than 30.

DS	Radius	N	$4D$ (308 m)			$5D$ (385 m)			$6D$ (462 m)			LX-SSA loss (%)
			LX-SSA	SSA	p	LX-SSA	SSA	p	LX-SSA	SSA	p	
1	500	4	56078.8	56086.8	0.237	56082.4	56085.2	0.700	56075.5	56083.9	0.839	0.19 → 0.19
1	750	8	111374.0	111506.5	0.245	111154.1	111163.4	0.871	110314.4 ²⁸	110847.5	0.021	0.88 → 1.85
1	1000	8	111924.6	111947.1	0.887	111953.5	111946.4	0.777	111781.5	111964.7	0.005	0.39 → 0.52
2	500	4	29004.2	29028.7	0.146	28999.5	29019.6	0.229	28981.8	28971.1	0.612	0.88 → 0.96
2	750	8	56659.2	56806.2	0.061	56621.0	56673.3	0.584	56533.9 ²⁸	56690.6	0.038	3.18 → 3.47
2	1000	8	57645.1	57739.3	0.084	57580.7	57655.7	0.140	57475.5	57587.2	0.114	1.50 → 1.79

X. SCOPE, LIMITATIONS, AND VALIDITY OF THE EVIDENCE

The results should be interpreted within the following boundaries:

- **Wake-model fidelity:** Jensen's wake model is retained for benchmark comparability and computational efficiency. Gaussian, CFD-based and data-driven wake models represent wake physics more accurately [34]–[36]. Re-evaluation with a Gaussian model changes the within-case orderings substantially (Section VIII-A); layouts were not re-optimized with it.
- **Power curve:** the piecewise-linear benchmark curve overstates expected power by about 18% (Data Set I) and 37% (Data Set II) relative to a cubic curve, so absolute energy values are benchmark values, not AEP predictions. The algorithm ordering is insensitive to this choice.
- **Spacing and boundary handling:** $4D$ is a legacy benchmark constraint; larger spacing reduces the constructible turbine count and changes the optimized layouts (Section VIII-B). The boundary-handling rule also affects absolute results and is held fixed across methods.
- **Budget and farm size:** all methods were compared at 6,030 objective calls. Larger budgets could change the ordering. At realistic farm sizes (80 turbines) the penalty-based methods did not find feasible layouts at this budget.
- **Algorithm settings:** the LX-SSA parameters ($\phi = 0$, $\chi = 1$) and the settings of the other methods are fixed defaults and were not tuned. Pseudo-gradient and surrogate-assisted methods are discussed but not benchmarked.
- **Fixed turbine count and engineering scope:** N is prescribed in every run. Terrain, heterogeneous turbines, electrical collector systems, grid, acoustic and environmental constraints, and economic objectives are not modeled.

XI. CONCLUSION

This work applies the previously published Laplacian Salp Swarm Algorithm (LX-SSA) [30] to continuous WFLOP. LX-SSA was developed in 2023 and is not proposed, redesigned or claimed as a new algorithm here. The contribution is an application study: the continuous coordinate encoding, the wake-coupled objective with explicit constraint handling, and a controlled, statistically analyzed comparison with five alternative methods on the standard Jensen–Weibull benchmark and on the Horns Rev 1 site.

In 68 benchmark cases and 12,240 seed-paired runs at equal budgets, the comparison gives five main findings:

- VNS is the best method, with an average rank of 1.60 and the highest mean in 44 cases.
- SSA, PSO and MS-SLSQP follow, and LX-SSA ranks fifth (3.97), ahead of DE.
- In the mechanism-level comparison with SSA, LX-SSA is never significantly better and is significantly worse in nine cases, so the Laplace follower candidate does not provide an equal-budget improvement on this benchmark.
- LX-SSA is clearly better than DE and more reliable than PSO and DE in reaching feasible layouts for dense farms, but less reliable than VNS and MS-SLSQP.
- At Horns Rev 1, with its real turbine, wind climate and outline, the same ordering appears. No method improves on the installed layout within the budget, and only the constraint-aware MS-SLSQP finds feasible layouts for the full 80-turbine farm.

Under the tested settings, LX-SSA is therefore not the preferred optimizer for this problem.

The algorithm ordering is insensitive to the power-curve model but changes considerably under a Gaussian wake model, and the absolute energy values depend strongly on both models. Further work should examine whether tuned Laplace parameters, a cheaper selection step or feasibility-preserving initialization improve LX-SSA, re-optimize layouts under higher-fidelity wake and power models, and study realistic farm sizes with correspondingly larger budgets and explicit constraint handling.

APPENDIX A

COORDINATES OF THE BEST LAYOUTS

Table XIV lists the turbine coordinates (m, farm center at the origin) of the best layout found by any method for the largest turbine count of each farm (Fig. 10). The final coordinates of all 12,240 runs, their convergence curves and the Horns Rev 1 layouts are provided as machine-readable files with the revision.

Compliance with Ethical Standards

Conflict of Interest: The authors declare that they have no conflict of interest.

Ethical Approval: This article does not contain any studies with human participants or animals performed by any of the authors.

TABLE XIV: Coordinates of the best layout for the largest turbine count of each farm (method and benchmark objective given).

DS	r (m)	N	Method	Objective	Coordinates (ρ_i, σ_i) (m)
1	500	10	MS-SLSQP	138171.2	(-315.2, 388.1); (277.9, -409.2); (-187.2, 43.6); (-27.6, -450.5) (-17.7, 306.6); (460.7, -160.4); (-365.7, -337.8); (-485.5, -33.0) (273.5, 408.8); (140.3, 40.7)
1	750	12	VNS	166626.2	(9.5, -749.9); (-255.6, 85.4); (574.3, -135.8); (134.3, 167.2) (-606.4, -345.2); (-116.6, -388.9); (297.6, -367.4); (687.2, 195.4) (-21.4, 692.3); (-743.2, 99.5); (-405.1, 546.3); (360.5, 625.1)
1	1000	15	VNS	208107.8	(769.0, 594.6); (978.8, -162.6); (11.0, 61.2); (-443.3, -12.6) (462.7, -101.3); (-137.6, -453.2); (-378.3, 424.2); (5.6, -883.2) (-849.4, 511.3); (646.3, 261.2); (-215.7, 874.6); (-486.1, -771.8) (466.4, 871.8); (-852.8, -403.4); (167.4, 600.7)
2	500	10	VNS	68162.3	(-11.9, -263.9); (472.0, -113.8); (184.1, 463.0); (-307.8, 131.6) (-211.8, 443.7); (99.2, 55.5); (291.5, -405.6); (-483.2, -126.0) (418.6, 201.3); (-294.9, -399.6)
2	750	12	VNS	83596.1	(60.6, 704.2); (136.6, -451.9); (-376.4, 634.2); (166.4, 296.3) (356.2, 559.5); (405.3, -631.0); (617.3, 15.1); (446.4, -249.2) (-722.1, 48.2); (-298.6, -220.1); (-435.5, -521.4); (-526.1, 326.1)
2	1000	15	MS-SLSQP	105183.7	(480.1, 134.7); (887.9, -308.8); (-932.2, -356.9); (-954.1, 299.4) (-999.9, -11.2); (469.8, 644.1); (-777.1, 625.3); (-106.5, 301.6) (-155.3, 748.4); (-430.9, -587.6); (-311.4, -949.7); (506.2, -845.8) (61.2, 997.2); (990.7, -16.0); (563.2, -540.5)

Informed Consent: Not applicable.

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